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Formal Project Proposal

Project Title:

PillPro's, Intelligent Medicine Dispenser

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TABLE OF CONTENTS

TABLE OF CONTENTS.....	II
LIST OF FIGURES.....	IV
LIST OF TABLES.....	V
LIST OF ABBREVIATIONS	VI
LIST OF EQUATION	VI
ABSTRACT.....	VII
1.0 INTRODUCTION	1
1.1 OBJECTIVE OF THE PROJECT.....	1
1.2 LIMITATIONS OF THE PROJECT	2
2.0 PROJECT SUMMARY	5
3.0 PROJECT BACKGROUND.....	11
4.0 LITERATURE REVIEW.....	13
5.0 PROJECT GOALS.....	15
6.0 METHODOLOGY AND PROJECT DESCRIPTION.....	16
7.0 PROJECT SCHEDULING BREAKDOWN AND TASKS.....	21
8.0 PROJECT IMPLEMENTATION	24
9.0 PROFESSIONAL COMPONENTS.....	32
9.1 ECONOMIC CONCERNS	32
9.2 ENVIRONMENTAL CONCERNS	33
9.3 SUSTAINABILITY CONCERNS.....	35
9.4 MANUFACTURABILITY CONCERNS	37
9.5 ETHICAL CONCERNS.....	38
9.6 HEALTH AND SAFETY CONCERNS	40
9.7 SOCIAL CONCERNS.....	42
9.8 LEGAL OR POLITICAL CONCERNS	44
10.0 STANDARDS.....	47
11.0 MOBILE APP	49
12.0 RESULTS.....	52
13.0 DELIVERABLES	60
14.0 FAILED IDEAS.....	62
15.0 DISCUSSION.....	65

16.0	CONCLUSIONS.....	67
17.0	REFERENCES.....	70
	APPENDICES.....	71
	APPENDIX A. ARDUINO CODE FOR PROJECT LEADER CODE:	1
	APPENDIX B. ARDUINO CODE FOR PROJECT FOLLOWER CODE:	19
	APPENDIX C. ARDUINO CODE FOR PROJECT: ARDUINO_SECRETS.H	28
	APPENDIX D. CODE OF ETHICS FOR ENGINEERING: IEEE AND NSPE	29
	APPENDIX E: USER MANUAL	31

LIST OF FIGURES

Figure 1. RFID Connection.....	7
Figure 2. Micro Servo motor	8
Figure 3. Leaser Light Sensor	8
Figure 4. Arduino Uno R4	9
Figure 5. Arduino Uno WiFi.....	9
Figure 6. DC motor speed Controller	10
Figure 7. Gantt chart for project schedule	21
Figure 8. Wire Diagram for all system	26
Figure 9. Block Diagram for all system.....	27
Figure 10. Programing Flow Diagram	28
Figure 11. Rectifier Circuit in Multisim.	31
Figure 12. Oscilloscope Measurement of Rectification Circuit.....	31
Figure 13. App layout.	50
Figure 14. Side View of the project	55
Figure 15. Front View of the Project	56
Figure 16. Initial prototype for pill holder	62
Figure 17. Second prototyped idea	63

LIST OF TABLES

Table 1. Project Parts and Prices.....	23
Table 2. Test 1 Result of the full assembled project.....	56
Table 3. Test 2 Result of the full assembled project.....	57
Table 4. Test 3 Result of the full assembled project.....	57
Table 5. Test 4 Result of the full assembled project.....	57
Table 6. Test 5 Result of the full assembled project.....	58
Table 7. Test 6 Result of the full assembled project.....	58
Table 8. Test 7 Result of the full assembled project.....	58
Table 9. Test 8 Result of the full assembled project.....	59
Table 10. Test 9 Result of the full assembled project.....	59
Table 11. Test 10 Result of the full assembled project.....	59

LIST OF ABBREVIATIONS

- WHO: World Health Organization
- RFID: Radio Frequency Identification
- HIPAA: Health Insurance Portability and Accountability Act
- FDA: Food and Drug Administration
- AC: Alternating current
- DC: Direct current
- PCB: Printed Circuit Board.
- IEEE: Institute of Electrical and Electronics Engineers
- VA: Volt Amperes

LIST OF EQUATION

- $V = I * R$
- $P = V * I$
- KCL: $\sum I_{in} = \sum I_{out}$
- KVL: $\sum V_{in} = \sum V_{out}$

Region	NMOS	PMOS
Cutoff	$V_{GS} < V_{TN}$ $I_D = 0$	$ V_{GS} < V_{TP} $ $I_D = 0$
Triode	$V_{GS} \geq V_{TN}$ $V_{DS} < V_{GS} - V_{TN}$ $I_D = K_n \left(V_{GS} - V_{TN} - \frac{V_{DS}}{2} \right) V_{DS}$	$ V_{GS} \geq V_{TP} $ $ V_{DS} < V_{GS} - V_{TP} $ $I_D = K_p \left(V_{GS} - V_{TP} - \frac{ V_{DS} }{2} \right) V_{DS} $
Saturation	$V_{GS} \geq V_{TN}$ $V_{DS} \geq V_{GS} - V_{TN}$ $I_D = \frac{K_n}{2} (V_{GS} - V_{TN})^2$	$ V_{GS} \geq V_{TP} $ $ V_{DS} \geq V_{GS} - V_{TP} $ $I_D = \frac{K_p}{2} (V_{GS} - V_{TP})^2$

ABSTRACT

PillPro's is a state-of-the-art, automated medication dispensing system designed to realize improvement in patient safety, increased medication adherence, and efficiency across clinical and home settings. This innovative system includes four different hoppers that can store various medications and a precise dosage mechanism driven by a combination of servo and DC motors. It recognizes patients with the help of RFID card technology, and hence this allows the system to give medication on the name of a particular person only after authentication and identification of the user. This feature reduces the risks of medication errors since this system ensures that at the correct time, proper medication is provided to the patient.

The core of the PillPro's system is a user-friendly touchscreen interface that allows users to interact with it in an intuitively understandable and intuitive way. The interface will guide the user through every step of medication intake by giving clear instructions and immediate feedback in real-time. It also includes vital information such as medication schedules, dosage details, alerts for missed or upcoming doses. Consequently, this function not only encourages adherence to prescribed treatments but also allows the patient to manage their own medication independently and confidently.

The mobile application that will be integrated into the system will give patients and their relatives added reinforcement. Thus, the system may allow access to data remotely by family members or caregivers interested in finding out the current state of the patient concerning the schedule of medications and adherence to them. The app allows for reviewing detailed information on dosage history, quantity of the medicine in stock, and notifications on missed doses. The app also provides the facility for updates when new medication is added into the hoppers so that at any time, the system remains updated with how much inventory of each medication is on hand and what dosing is required. This is very helpful for caregivers who may have patients in different stages with complex medication regimens for either elderly or chronically sick patients.

PillPro's is designed with scalability and adaptability, making it suitable for applications ranging from individual patient care to large health care facilities. The system is modularly designed so it can be easily adapted and tailored to fit various medication regimens and patient

needs. Advanced data logging and reporting features will also be available through PillPro's, enabling health care providers with access to detailed records of adherence and usage patterns. This in turn will be useful for informed decision-making on patient care, modification of treatment plans, and developing strategies to handle challenges in medication compliance.

Automation of medication dispensing by PillPro's will decrease the administrative burden on healthcare professionals and free their efforts for direct patient care. It will also drastically cut down the chances of medication errors, one of the most prevalent causes of adverse events in healthcare settings. From automated dispensing, through ease of use in design, to extensive remote monitoring, the PillPro's are positioned to revolutionize medication management. This system supports not only patients in their compliance but also enhances the quality of care through real-time information and proactive intervention.

From the hospitals and nursing homes down to single-family households, the reach of PillPro's may unleash a transformative revolution in medication management approaches across different levels of health settings. Consequently, PillPro's shall contribute to the realization of improved health outcomes and quality of life through optimized safe and effective medication use, improved adherence, and communication between patients, caregivers, and health providers.

1.0 INTRODUCTION

1.1 Objective of the Project

The PillPro's endeavor would be about changing the face of medication management for administration, an automated system that would aid increase patient safety, adherence to medication. The main focus would be to develop a dependable and easy-to-use medication dispensing system, enabling proper and timely delivery of prescribed medications to the patient, with maximum support from the family members or caregivers.

Among the key objectives of PillPro's is to improve adherence to medication, an integral component of meeting successful health results. Poor adherence to prescribed regimes can have grave consequences for poor health outcomes: it may increase illness exacerbation and hospitalization, thus raising healthcare costs. In that respect, PillPro's automates the dispensing process and reminds on time with its integrated touchscreen interface and mobile application. This system reduces the possibility of missed doses, hence allowing the patient to keep close to their medication schedule and contributing to better management of their health.

Besides promoting compliance, PillPro's has been designed in such a way that it significantly reduces medication errors, one of the serious concerns across healthcare settings. Such medication is delivered to the patients after authentication and identification through RFID technology. It safeguards that the right medication is given to the right person at the right time. This degree of precision avoids risks of such mistakes as wrong medication and wrong dosages, which may cause serious and sometimes life-threatening consequences for the patients.

Another key objective of the PillPro's project is to facilitate remote monitoring and participation in care by family or any other caregivers. It enables authorized users to view real-time medication history, dosage schedules, and current inventory levels of the patient through its companion mobile application. This is especially helpful for families that have to take care of either elderly or chronically ill patients, as it keeps them in the know and involved in the patient's health care. Additionally, the mobile app will provide an avenue through which users can update the system if there are additions of new medication to the hoppers, hence the device will actively update itself for timely and correct dispensing.

Another important mission of PillPro's is to facilitate the process of medication management for healthcare providers in general. Data logging and reporting within the system will be insightful regarding patient adherence and usage patterns. Healthcare professionals will have access to this information in order to monitor compliance, make informed decisions regarding treatment options, and intervene proactively, if any issues are identified. The medication management system would not only make direct patients' outcomes better but also lighten the administrative workload of medical personnel in the process. This lets them have more time for direct patient care.

Safety and quality of care are at the center of the project for PillPro's. Automation of the medication-dispensing progression reduces the risk of human error with the system while assuring the delivery of prescribed medications in a safe and consistent manner. It is easily accessible and user-friendly for both the patient and caregiver due to its friendly design of the device and integration with a mobile application. If applied clinically or in a home setting, PillPro's has great potential to revolutionize medication management practice since it would be so much more effective, accurate, and truly centered on the patient.

The key objective of the PillPro's project is the development of an automated, innovative medication dispenser to significantly improve medication adherence and decrease incidents and errors associated with it, while enabling remote monitoring. By integrating state-of-the-art technologies with user-centered design, PillPro's aims to guarantee patient safety, better involvement of caregivers, and contribute to improving health outcomes and the quality of life in the patients.

1.2 Limitations of the Project

The automation of medication dispensing involves the integration of RFID, servo and DC motors, a touchscreen interface, and multiple hoppers for holding different types of medications within the PillPro's Project. While it offers great promises for efficiency in medication management, there are some limitations that must be recognized to ensure proper implementation and satisfaction from its users.

The major limitation provides insight into the RFID technology utilized. In as much as RFID provides an efficient way of authenticating users and dispensing prescription medication, it has a

limited operating range of up to 10 meters. This can limit its usability in larger areas where users may be required to stand some meters away from the dispenser. RF signals can also be interfered with by any metallic or electronic device present in the environment; this could make the system less reliable. High-quality RFID tags may be expensive, and hence, the cost factor may give a dent to the whole project budget.

While the touchscreen interface increased users' interactions, it is not without its drawbacks. It works only to the extent that it is designed well. A poorly designed user interface might cause user errors and frustrations. Thirdly, mishaps such as spills, scratches, or heavy use might damage these touchscreens hence leading to costly replacements or repairs. Relatively often, calibration might be required for accurate touch detection, further adding to the maintenance hassle.

Additional challenges are created by the mechanical parts of the PillPro's, essentially the servo and DC motors. This can further cause mechanical malfunction in these motors, which can lead to malfunction in the system's dispensing mechanism and generally cause disruption of service. The noise from the motors' operations can be disruptive, especially for quiet environments; this may discourage users from using the system in such environments. Also, the dependency on a constant power source reduces the degree of portability because using a battery backup would further increase complexity and reduce its overall weight, which may not be feasible or desirable for all users.

Medication storage and management is another pivotal point. The PillPro's system has limited capacity in each hopper, requiring frequent refills besides proper inventory management to ensure that medications will be available upon request. Besides, different medications usually have their peculiarities in storage conditions regarding temperature and humidity, which may complicate the design of the system. Another significant issue could be the implementation of an effective system for tracking expiration dates and notifying users about medications that have expired; this would require substantial software development.

Other things that stand in the way of success with PillPro's include user adoption and training. Users will have to get used to a more technologically savvy system than they may be used to in handling their medication. Non-tech-savvy people could take a great deal of training to get them up to speed and may discourage some users from fully embracing the new system.

There is likely also to be some resistance to change with those used to other, more traditional forms of medication management.

Finally, the PillPro's project must take utmost priority when it comes to safety and security. From storing personal and pharmaceutical information, huge concerns regarding data privacy arise, which require protective measures for sensitive user data at all costs. Moreover, medication dispensed in error may lead to serious health consequences; thus, appropriate fail-safes and rigorous testing are highly important to ensure proper functionality of the system.

While PillPro's has been designed to represent a change in thinking in medication dispensing, there are several limitations that must be weighed and developed for functionality, usability, and safety. These will require thoughtful design, exhaustive testing, and user feedback continuously to make the project a total success. By recognizing yet working to mitigate these limitations, PillPro's can meet its charter of serving users and their families better in their medication management.

2.0 PROJECT SUMMARY

PillPro's is an innovative medication dispensing system specifically developed to improve medication management for people with dementia and other cognitive disabilities. With over 55 million people affected worldwide, and over 60% of those people living in low- and middle-income countries, the need for efficacious solutions that promote medication adherence and improve patient outcomes has never been so great. PillPro's addresses this very serious health concern by providing a quite user-friendly system that incorporates advanced technologies: RFID capabilities, an intuitive touchscreen interface, and a very sophisticated app for caregivers. This will make medication management easier.

The medication management system will include four types of specialized medication hoppers that store a number of different medications safely and securely, meeting the needs of these complicated and often complex medication regimes associated with dementia. Each hopper is programmable for a specific medication to be distributed at specific times, so that when the time is due, the patient will receive their dose of medication. By using RFID technology, the PillPro's can correctly identify the patient and then dispense the proper medication based on that particular individual's needs. Not only does this minimize the chances of missed doses or out-of-schedule doses, but it also reduces the risk of those common medication mistakes that would make all that difference in health and well-being for a vulnerable population.

Core to the experience of PillPro is an in-built touchscreen interface designed for ease of use by patients and caregivers alike. This interface makes for easy navigation, hence allowing people with cognitive impairments to interact well with the device. Caregivers, too, play a very important role in the management of medication, and this is put into consideration because PillPro's offers a companion app dedicated to the caregiver in order to monitor adherence in real time. With this app, a caregiver will be able to be alerted on every administration, view history of usage, and change medication schedules when appropriate. It empowers the caregivers and can ensure more communication between the patients and their networks of support for improved health outcomes.

Compared to other products available, including the Hero Automatic Medication Dispenser and the e-pill Voice, PillPro's will offer more integration and customization to each user's

specific needs. While these can be fantastic solutions for medication management, many have limitations that can make them less effective. The Hero does not allow for half pills or medications needing refrigeration, and these may be critical medications that cannot be used with this device. Other features, such as the use of a passcode, are for security but could also create challenges for users who have issues with memory and work hard to manage independently. While the e-pill Voice appropriately includes auditory and visual reminders, there is still a gap in terms of accessibility and ease of use.

PillPro's aspires to fill these gaps through design that this technology is accessible to the user and also has the caregiver involved. PillPro's does not have complicated access codes; it's just simple to use, thus making interactions easy and captivating for users who will be more independent in managing their medication routines. What's more, by this tailored customization, the schedule can be changed because every patient has a different need, so that is going to promote their independence and improve their quality of life.

The aspect of PillPro's is not just technological; it is very human. The system will foster a supportive environment for patients with cognitive impairment through the maintenance of dignity and independence, while ensuring that the level of care they deserve will be provided. In the course of its development, PillPro's integrates feedback both from the patient and the caregiver to create a solution that speaks directly to the unique challenges this population faces.

Conclusion: PillPro's is a new frontier in the field of medication management technologies and a system that is bound to bring a change in the lives of individuals suffering from dementia, as well as their caregivers. PillPro's enhances life quality for individuals living with cognitive impairments through the assurance of safety and effectiveness in medication management. Because conventional thinking holds that demands for innovation within the healthcare sector continue to grow, PillPro's literally stands at the forefront of addressing one of the most nagging health problems of our times.

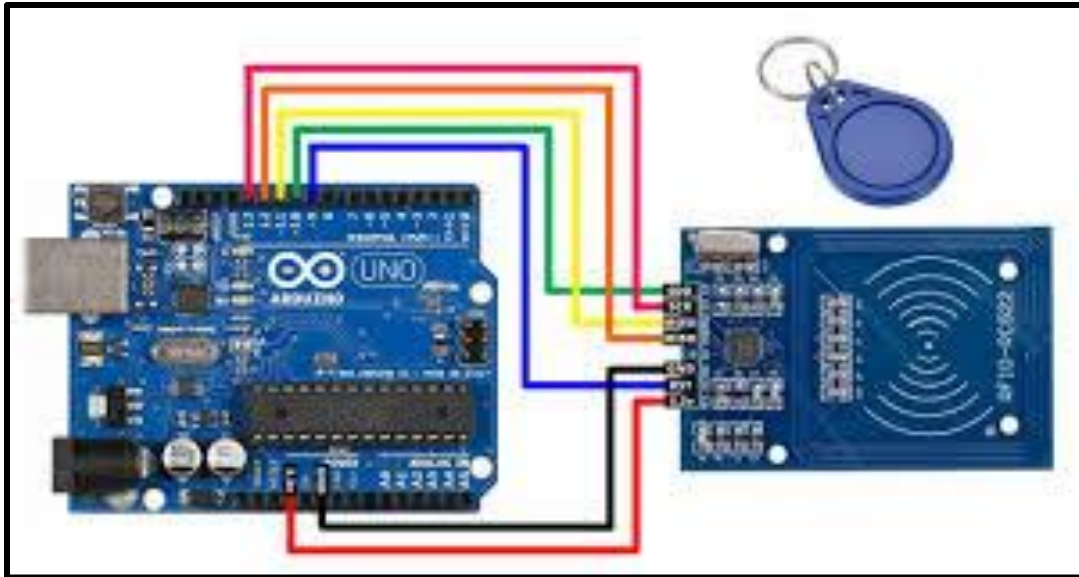


Figure 1. RFID Connection

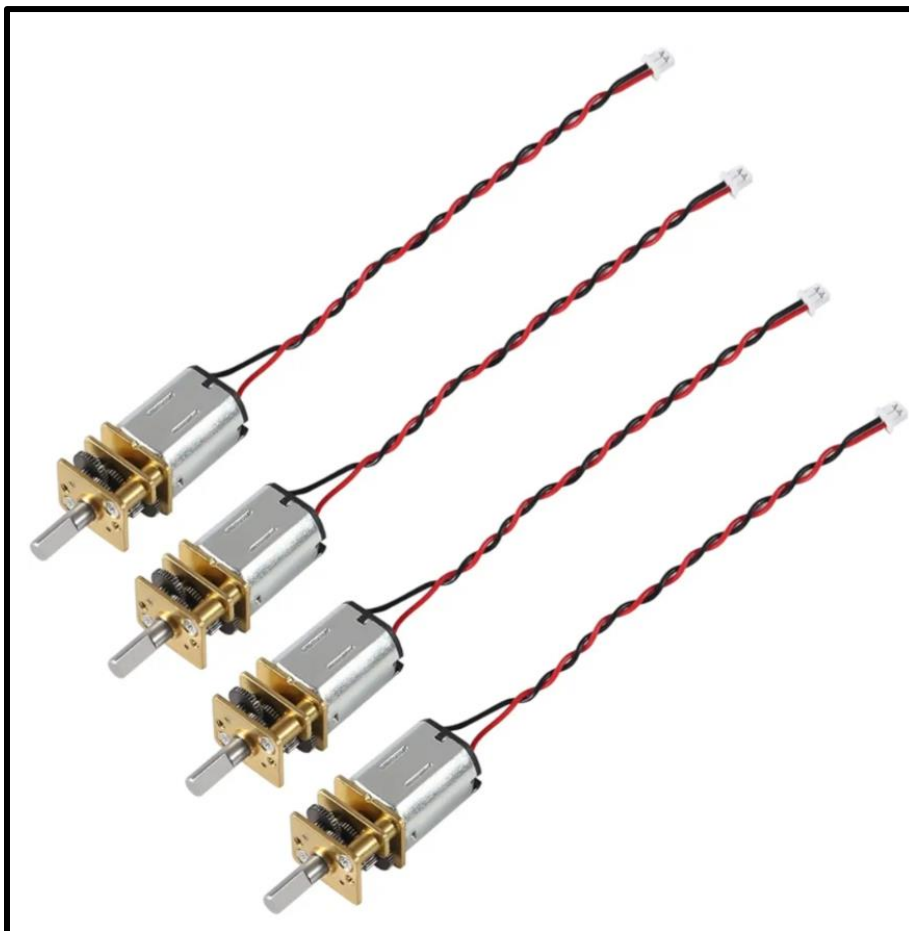


Figure 2. DC motor 3.3V

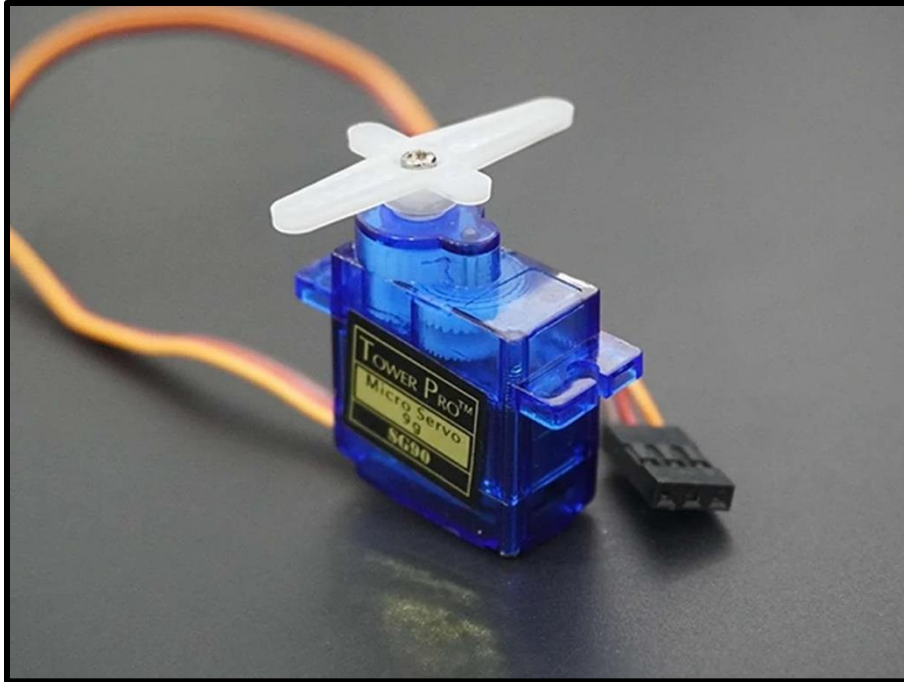


Figure 2. Micro Servo motor

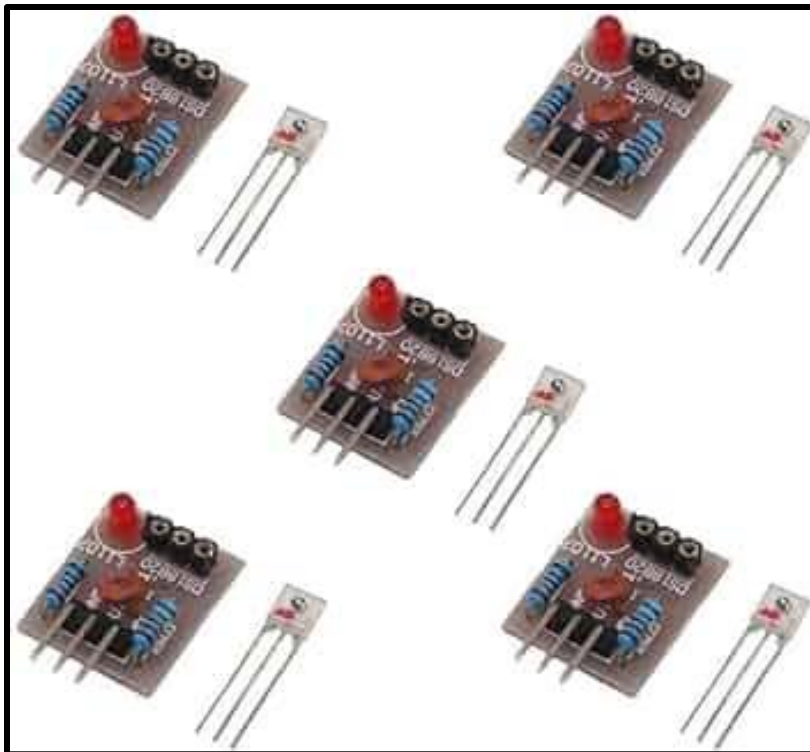


Figure 3. Leaser Light Sensor

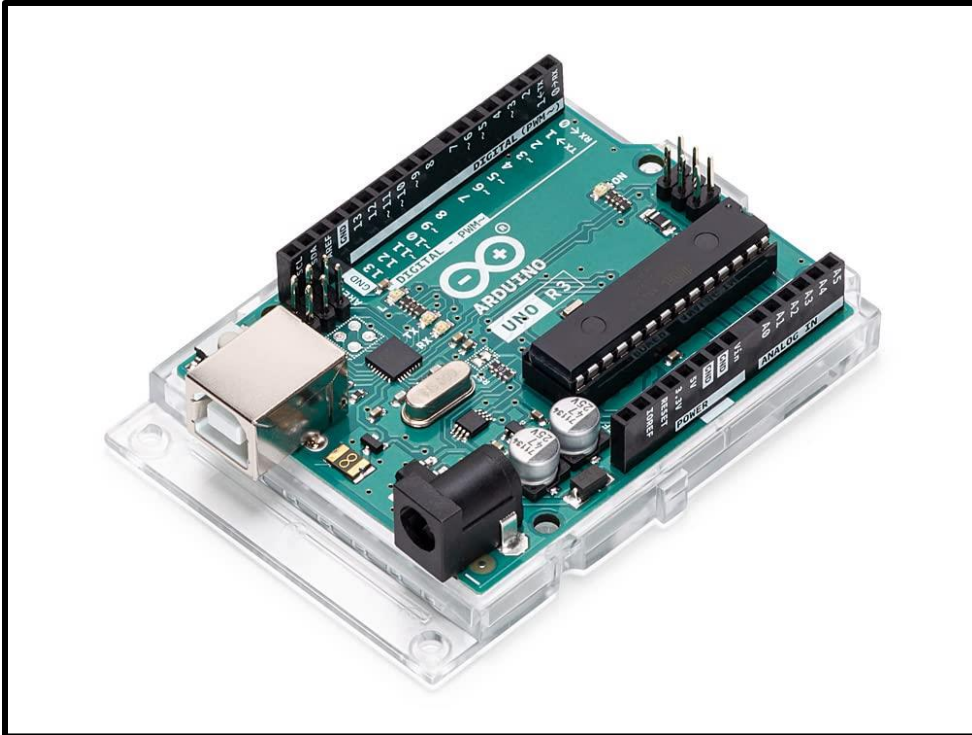


Figure 4. Arduino Uno R4

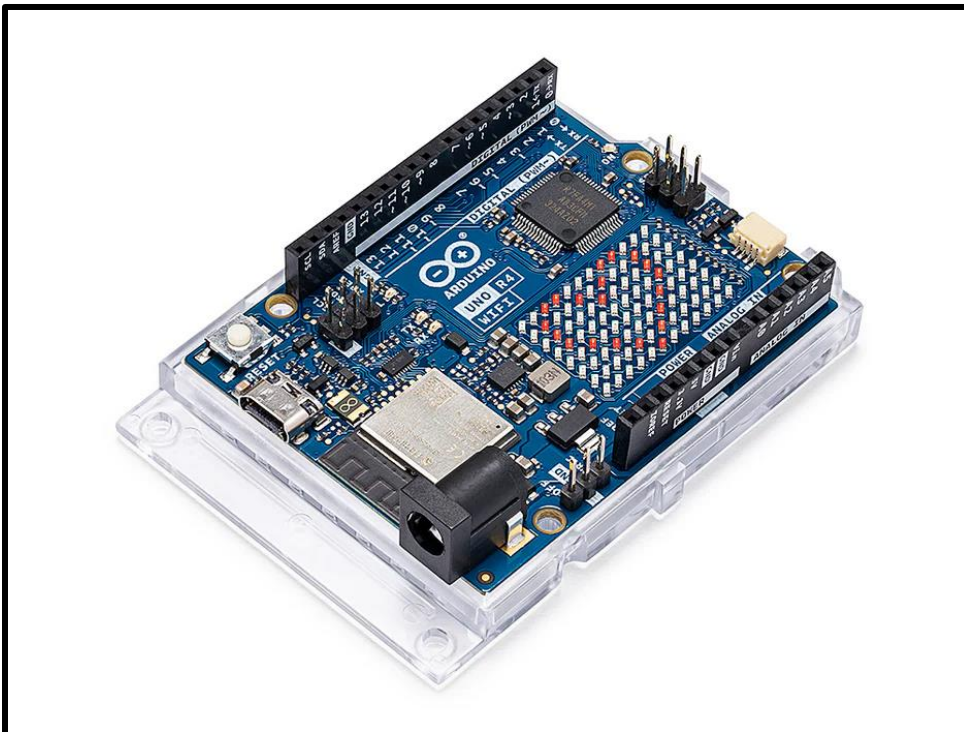


Figure 5. Arduino Uno WiFi

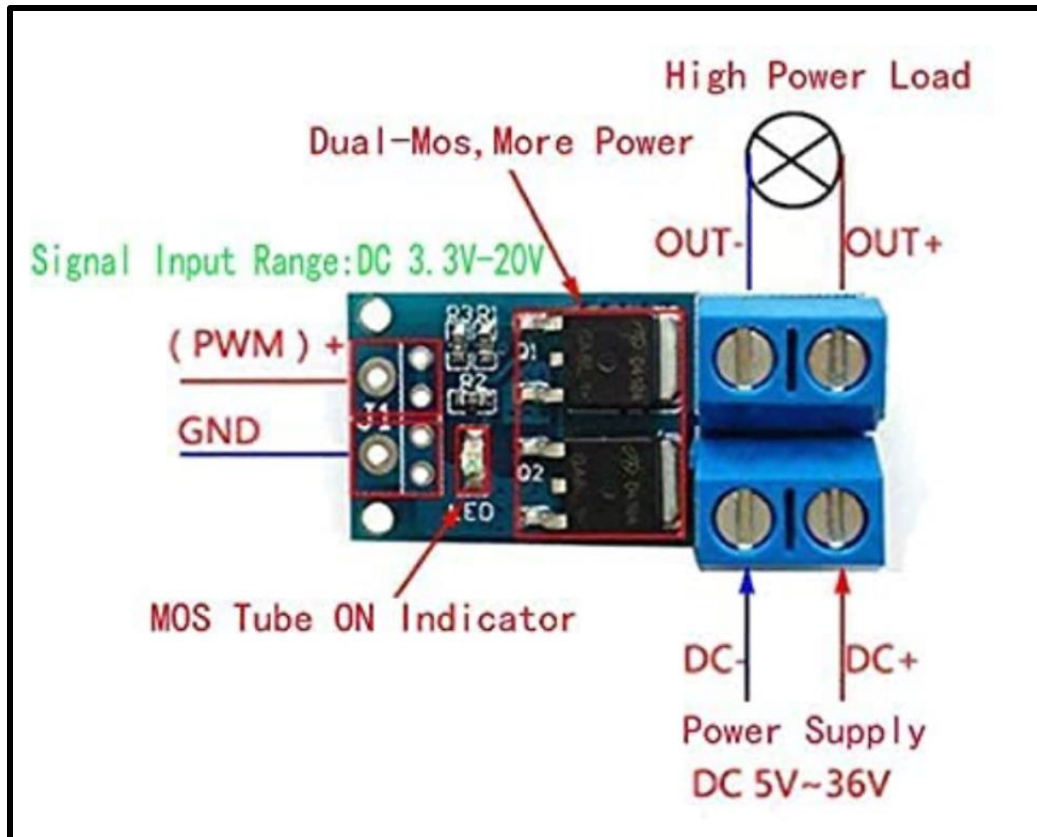


Figure 6. DC motor speed Controller

3.0 PROJECT BACKGROUND

Dementia is a neurological condition that already affects over 55 million people in the world, increasingly being recognized as one of the most important global health challenges of our times. According to the World Health Organization, more than 60% of people with dementia live in low- and middle-income countries where infrastructure for providing care can be at a minimum and resources insufficient. This has contributed to the increasing burden of health systems globally with close to 10 million new cases diagnosed annually. Dementia results from different diseases and injuries that generally affect the brain, with the most prevalent being Alzheimer's disease, constituting 60-70% of the cases. Dementia is the seventh leading cause of death in the entire world and one of the major causes of impairment and dependency in elderly populations. Thus, finding a solution to the condition means managing highly complex challenges affecting the patients and their caregivers.

Dementia affects many aspects of a person's life. Of all these, probably the most complex one is medication management. The complicated medication regimes, often necessary to manage the symptoms of dementia, or other health conditions, overwhelm the patient so easily and lead to omitted doses, wrong time of intake of medication, or wrong scheduling. These errors could certainly worsen the health problems of the patients, and hence, effective medication management solutions form an important part of dementia care. The development of such innovative tools as smart pill dispensers, which could help the patient stay on schedule and provide much-enhanced compliance and safety.

With this increasing demand for dependable medication management, there appear to be quite a number of smart pill dispensers that seem to be the game-changers in the market. The Hero Automatic Medication Dispenser is one of the most well-known devices on the market. This is one of the most prestigious devices with a set of smart features to improve adherence. Hero is capable of managing up to 10 different medications stored in it or managed through its companion app for flexibility with complex regimens. It can hold a 90-day supply of medication, reducing frequent refills, and it accommodates a range of multiple daily dosing schedules, which makes it very suitable for patients with complicated needs. Hero also has some basic safety features, such as a passcode lock to prevent unauthorized access, to make sure only those who actually should handle the medications can. Despite all these various benefits, Hero does have a

few limitations. It does not support half pills or refrigerated medications, which can be considered a drawback for users with such specific needs. This means that the high price of the device, plus subscription fees for its services, might be unaffordable to certain users. While this adds to security, it might be more of a challenge for users with memory impairments since it will not allow them to easily access their medications independently.

Another prominent product in the smart pill dispenser market is the e-pill Voice that does medication management in a slightly different way. This, as a result of the auditory and visual signals, assists a patient in taking their medication on time, whereby the automatic pill dispenser is locked. It provides six daily sets of alarms featuring a clear voice and blinking lights, alerting users when it's time to take medication. Especially useful for individuals with varying levels of cognitive function, the reminders are easy to understand and difficult to miss. The e-pill Voice is powered by an AC adapter or batteries, giving users flexibility in where and how they use the device. Its 28-compartment medication tray and key-locked lid provide both convenience and security, helping to prevent tampering or accidental overdose. Similar to Hero, though, the e-pill Voice also is relatively expensive, thus limiting access for some users. That said, its focus on auditory and visual reminders makes it extremely useful for the more vulnerable users, especially of its class.

While Hero and e-pill Voice boast features for easier medication management, PillPro's hopes to build on these innovations by providing a more integrated and user-friendly experience. The use of RFID technology in concert with an easy-to-use caregiver app overcomes some of the weaknesses inherent in other devices. Passcode use, for instance, is cumbersome, especially for those suffering from memory defects; this is offset in the approach taken by PillPro's. Rather than requiring manual input, the device enables caregivers to remotely monitor and manage medication schedules so patients can maintain their independence without sacrificing safety. More personalization and ease of operation will make the device highly attractive for those patients who require further personalization with extended support and engagement from caregivers. The device can be adjusted to meet very specific patient needs, thereby making operation easy and further situating it within the growing smart pill dispenser market. With dementia care solutions in increasing demand, it puts PillPro's in a very promising place to improve medication adherence, safety, and quality of life in both patients and their caregivers.

4.0 LITERATURE REVIEW

The PillPro's project represents a major step forward in the field of technology related to medication management, in an effort to increase the accuracy and efficiency of the distribution of medication to patients, thereby eliminating most-if not all-of the many concerns associated with its use. The interaction between ethical, health and safety, social, legal, and political considerations underline the difficulty of developing a system that would be able not only to match the expectations set forth by the technologies involved but also to meet diverse user needs and comply with regulatory requirements.

Central to the PillPro's project is the commitment to uphold ethical standards in safeguarding user health and privacy. Following the IEEE Code of Ethics, the projects underline the importance of integrity and responsible behavior during design and implementation. Ensuring that the system considers users' safety is not only a legal requirement but an ethical one; medication errors can result in life-altering consequences. This commitment to safety is inextricably entwined with health and safety concerns; hence, a strong test and validation process will be required to be assured that the project minimizes risks associated with medication dispensing. The requirement for tight quality control and ease-of-use interfaces further reflects the commitment of the project to a safe, accessible environment for all users, but in particular the vulnerable elderly or those with various disabilities.

Social issues also form part of the successful development of the PillPro's system. Basic access and equity will determine whether all persons will be able to benefit from the capability of the project despite their status in terms of socioeconomic and technological literacy. During the development phase, community engagement develops a sense of ownership and relevance; consequently, the public health outcomes increase. Besides that, privacy handling and data security require not only compliance with the law but also trust-building within the community. These are social issues, one leading to the other, which indicate how such user involvement, from concept to through the life of the project, demands transparency.

Legal and political considerations further complicate the landscape for PillPro's. The legitimacy and safety basis for the system remains with healthcare regulations imposed, for example, by the FDA and HIPAA. However, navigating the issues of liability, intellectual

property rights, and regional legal variations is considerably complicated. Active steps will be taken to understand and address these legal frameworks as an important mitigant to ensure the viability of the project in various jurisdictions. Engagement on such fronts with policymakers and stakeholders at so many levels might ease integration into the health system, bring the PillPro's project in line with current political priorities, and respond to community needs.

Of course, here lies the key to success-whether the project addresses all these concerns in a comprehensive and cohesive manner or not. The interplay between ethical, health, and safety; social; legal; and political issues calls for a consultative network that involves not only stakeholders but also health caregivers, community-based organizations, and regulating authorities. It is through this open dialogue and constant assessment that the PillPro's team will find easy accommodation of emerging challenges and opportunities to keep the system relevant, effective, and helpful in securing better medication management and health outcomes.

In fact, the PillPro's project is an approach toward innovation with ethical responsibility and community involvement. The project will be able to make a much more significant contribution to the healthcare environment with the myriad issues accompanying the design and implementation of medication management technology. By building robust safety measures into our architecture, considering mechanisms of accessibility and equity, and following the legal and ethical standards within the field, PillPro's will be well-placed to play an important role both in the improvement of medication adherence and, generally, in improving patient outcomes.

5.0 PROJECT GOALS

The main aim of the project is to design and construct a multi-user household medicine dispenser with efficiency and ease of use in mind. Considering that the PillPro is targeted for people with Alzheimer's disease, head injuries, or memory loss, some kind of system needs to be developed whereby the caregiver can monitor dispensing and have immediate notifications when refills are needed. That is where PillPro differs from the conventional medication dispensers. Our innovative device will integrate flawlessly with a smartphone application, therefore making it accessible on any mobile device. The app will feature an interface that allows users to store crucial information, such as the patient's name, the medication details, and the exact dosage to be given. This increased connectivity and monitoring capability will ensure proper administration of the right medication at the right time, enhancing both safety and usability for both caregivers and patients.

Its second aim is to be able to facilitate a variety of medications of unique shapes and sizes with ease. Medications can range from small-sized round tablets to larger-sized elongated pills, and therefore PillPro has to be able to dispense each accurately, such that only a single pill would come out at a time. Another very important consideration is that many, many patients have to take more than one medication either all together at one time or at multiple times throughout the day, and therefore, the device should be able to accommodate a minimum of three types of medication. Such versatility is highly critical to being able to address various treatment programs with convenience and accuracy for the patient and caregiver alike.

The third aim of the project is to ensure everything is 3D printed or sourced from readily available parts. In this way, not only is the process for replacement when needed much easier over the lifetime of the machine, but it also keeps costs as low as possible and makes maintenance as painless as possible. We can keep the production of these in-house by 3D printing the main components. In that way, dependency on third-party companies will be reduced, which will make the manufacturing process go faster and reduce waiting times for replacement parts. On the whole, it will quicken the production timeline. Continuous design improvement will also be enabled, together with rapid prototyping, by in-house 3D printing, assuring the product continuously evolves toward fitting future needs.

6.0 METHODOLOGY AND PROJECT DESCRIPTION.

The PillPro Smart Pill Dispenser is a mechanized medication dispenser that helps people remember to take their medication at the right time. This automated device greatly helps a person who suffers from chronic ailments or cognitive diseases like Alzheimer's disease, memory loss, or any other disorders that may lead to not remembering to take their medication on time or as prescribed. The dispenser makes use of RFID technology that identifies users, dispensing the correct medication on schedule, considering both safety and convenience.

PillPro can handle multiple medications, issuing out the proper dosage according to the programmed schedule. In addition to the core function of dispensing, there is a user-friendly touchscreen interface for easy setup and real-time adjustment of medication schedules. This device will also be integrated with a mobile application that allows caregivers to monitor remotely the medication adherence of their patients. The app provides alerts on low medication levels and missed doses, among other critical updates, for an all-inclusive care solution.

The main task of the PillPro is to minimize the risks of missed doses, overdose, or incorrect medication, which would provide significant improvement in user quality of life and indirectly decrease the burden on their caregivers. The apparatus should be straightforward and user-friendly to be operational at home and clinical settings.

Methodology

The design of the PillPro Smart Pill Dispenser follows a structured and iterative approach. This project is to be divided into distinct phases of design, prototyping, software development, integration, testing and validation. Specific details for each phase will be reviewed in the sections below.

1. Design Phase

First is the conceptualization and design of this smart pill dispenser to meet the needs of the users regarding how the device should work and how it integrates into the overall system. Some of the major design decisions were made during this phase to ensure that the device will meet its objectives in terms of accuracy, automation, and friendliness to users.

- System Architecture Design: The PillPro is based on an Arduino platform due to simplicity, flexibility, and great community support. The architecture involves:
- Arduino Uno R4 WiFi controls the main functions of the system.
- RFID Reader identifies the user scanning their own RFID tags.
- Servo motors control the pill dispensing mechanism.
- DC motor to feed medications into a tube in the dispenser for refills.
- Touch screen interface, so that users and caretakers can operate the entire system with ease.
- WiFi connectivity to connect it with the mobile application.

By the end of this stage, a block diagram had been developed that mapped all the preceding connections between the various components. Circuit designing and prototyping on a breadboard allowed for the testing of each component, to be assured that it works properly and will integrate into the final build.

2. Prototyping Phase

Once the design of the system had been finalized, the next step was prototype working. The different hardware components were assembled and wired in accordance with the designed circuit. In this project, the third phase of mainly focused on:

Hardware Assembly: Wiring of the RFID reader, servo motors, DC motors, and touchscreen interface with the Arduino Uno R4 was involved. Other components used include power supplies and communication modules, which were integrated to ensure that the system would work reliably over a long period.

Prototyping of Pill Dispensing Mechanism: The pill dispensing mechanism was designed in a way that it dropped the pills inside a slot according to scheduled dosage. This was all controlled through the servo motors, which were programmed to rotate the pill container and let the pills out according to RFID input.

It was used to print dispenser parts that housed the pills and mechanisms. Plexiglass was used for covers where transparency was necessary for visual monitoring of the levels of pills without necessarily opening up the system.

3. Software Development Phase

Arduino Code Development

The Arduino integrated development environment was used to design the software that controlled the PillPro system. The RFID library was added along with servo motor control and touchscreen interface libraries for controlling the different components of the gadget. Then, the gadget's software was supposed to execute the following tasks in order:

- Reading RFID tags to identify users and match them to their specific medication schedules.
- Dispensing medications via servo and DC motors based on user input and time schedules.
- Presenting information, such as forthcoming doses, present levels of pills, and the status of the system on the touchscreen.

Development of Mobile Application:

The application has been developed on the mobile devices of caregivers for remotely monitoring the pill dispenser. This mobile device connects to the PillPro system through WiFi and provides the following functionality:

- Real-time updates about the status of medication and any other dispensing issues.
- Refill alerts of medication when the level of the same in a dispenser falls below a minimum limit.
- Storage and synchronization of historic data of the intake of medication.

This required that the mobile phone application be very carefully designed so that it communicated expressly with the WiFi system on the Arduino.

4. Integration Phase

After hardware and software development, integration between the two was the next procedure. This stage was strictly meant to ensure that the hardware components involved-the motors and sensors-are communicating appropriately with the software for any successful medication dispensed accurately.

Major Steps of Integration:

- The RFID reader was tested to see whether it identifies users properly and communicates data to the microcontroller.
- The servo motors were aligned with the software so that pills are dispensed only after reading a valid RFID tag.
- The touchscreen interface was then interfaced with the Arduino code so that the schedules are shown along with the user input and any errors.
- WiFi was also tested to be in working order to ensure proper communication between the mobile application and the dispenser.

5. Testing and Validation Phase

Testing was an integral part of the development process in ensuring that the PillPro system works both reliably and effectively in real-world use. The key methods employed during the system testing were as follows:

1. **Software Simulations:** Software simulations were performed to test whether the logics controlling the pill dispensing, recognition of input through RFID, and user interface interactions were performing as expected. This step helped find any bug in the software that could disrupt the functionality of the system.

2. **Real-World Testing:** The prototype has been put through various practical tests to affirm the accurateness of the pill-dispensing mechanism. Multiple users with multiple RFID tags were tested to ensure the system would correctly dispense medications according to the schedule of each individual user. Some of these tests included:

- **Dispensing accuracy:** Whether the correct dose will be delivered every time.
- **System Response:** The time of RFID scanning and dispensing can be achieved within the tolerance.
- **Touchscreen Usability:** Any users input or adjust their schedules can be made with much ease.
- This had to undergo several rounds of optimization through testing to achieve perfection in accuracy and reliability.

6. Data Collection and Optimization

Data Collection:

In the course of testing, data was collected on each user's medicinal schedule, precision in dispensing, and adherence. The data is recorded on the microcontroller and then sent to the mobile application for data analysis. It is at this point important that this should be able to be viewed remotely by caregivers so that they can make any adjustments to the system when necessary.

Optimization: From this gathered data, several enhancements were made to improve the overall performance of the system. These include fine-tuning in the mechanism of dispensing to avoid any jamming once pills are released and updating the software for easier interaction of users on the touchscreen. Refining the interface of the app is also done to afford more intuitive navigation by the caregivers.

7. Final Validation

After testing and optimization of all the components, the final validation of the PillPro system was done. This ensured the device met all the initial project objectives: to dispense medication with accuracy and on time, to integrate well with RFID tags, to ensure reliable communication between the mobile application and the dispenser, and to provide an interface that was user-friendly and could be well manipulated by both users and caregivers.

Conclusion

PillPro is a smart pill dispenser in the holistic solution for people in need of medication. The project solves the challenges posed by the emerging need for accurate and reliable dispensing of medication by incorporating, into complex software development, hardware components involving RFID readers, servo motors, and touchscreens. Iterative design, prototyping, integration, and testing had to be performed to ensure that the final product was going to work functionally and also be friendly to the user. This ongoing testing and collection of data enable the optimization of the PillPro system, aiming at an improved quality of life for the users while minimizing the risk of medication errors and offering peace of mind to both patients and caregivers.

7.0 PROJECT SCHEDULING BREAKDOWN AND TASKS

Figure 1 is the Gantt chart that we followed for the scheduling of the project. This chart was able to set deadlines for certain tasks to be completed. This ensures that we stay on track with the time schedule that is permitted. Looking at the chart below in figure 1, We have ensured that ample time would be left for testing of the complete design. The main reason for this is because there are a lot of moving parts, and those parts need to move in an exact sequence in order for the machine to dispense the exact medication and the precise dosage. Most of the time for this project was spent on assembling all the parts together.

Pill Pros

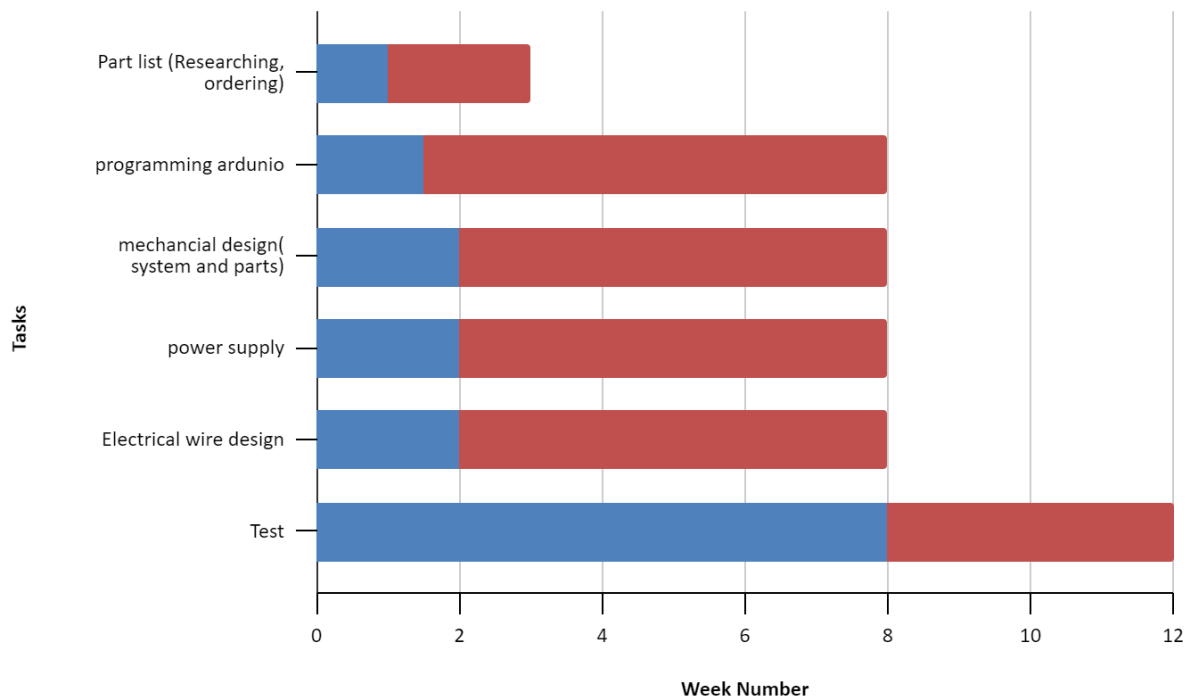


Figure 7. Gantt chart for project schedule

The parts list and research needed for the project will take approximately two weeks. The components used are standard for Arduino-based systems. However, collecting and reviewing datasheets to ensure compatibility is essential. It's also crucial to check the voltage and current requirements for each component to ensure longevity. Compiling a list of the different voltage

levels will enable us to plan how each component will be powered and determine the voltage requirements throughout the project.

Programming the machine will be the most time-consuming task, taking around seven weeks. The programming will evolve throughout the project as new ideas are implemented and testing begins. During the final testing phase, we will fine-tune the program code for controlling the motors, servos, laser sensors, and touch screen. This is necessary because the medicine dispenser needs to execute sequential steps with precise timing to maintain accuracy.

The mechanical design of the project is complex and will require multiple iterations to ensure compatibility with all components and prevent medication jams. Additionally, the design must guarantee that only one medication is dispensed at a time. Mechanical design is expected to take about six weeks to complete. Since all the mechanical components are being 3D-printed, each part must be carefully designed and tested to ensure compatibility and functionality with the selected components.

The power supply section of the project was designed and built from scratch using knowledge from a microelectronics course. It can be plugged into a standard American 120VAC wall outlet. The power is then routed into a custom 3D-printed case that houses the circuitry for power rectification. Safety measures, including a fuse in the AC-to-DC converter, have been added to ensure that the current does not exceed 5A.

The AC-to-DC converter also provided an opportunity to design a printed circuit board (PCB) for the project. Before proceeding with the PCB design, the circuit will first be constructed in Multisim. Multisim is beneficial as it allows us to simulate the circuit and verify the output voltage using an oscilloscope. Once the expected results are obtained, we will prototype the circuit on a breadboard to confirm its accuracy. After successful testing, we will use KiCad to design the final PCB.

The electrical wiring phase of the project is scheduled to take around six weeks. This extended duration is due to the sequential assembly of components within the case. Ensuring that each component is correctly connected to its respective pin location on the Arduinos is critical for testing. Our design follows a tiered structure and will require wiring from the bottom up.

Connectors will be used for the touchscreen and Arduinos to facilitate easy disconnection for component replacement or troubleshooting. KiCad will be used to create a wiring schematic that illustrates all components and their connections. This schematic will help minimize connection errors and simplify troubleshooting.

We have allocated four weeks for testing the medication dispenser. This time frame should be sufficient to identify and resolve any issues in the mechanical design, program, and component integration. It will also allow us to adjust the timing of the program code to fine-tune the control of the different components.

Reviewing table 1 will show the parts used in this project and the cost associated with it. We have tried to keep the project cost-effective to allow for this product to be replicated and produced for the general public for an inexpensive price. This will help us to be competitive in the market for pill dispensers.

Part	Quantity	Price
Micro Servo	8	\$17.99
Step Down Transformer	1	\$15.97
Laser Sensor Module	4	\$12.99
3V-6V Micro Motor	4	\$10.99
Arduino Uno R4 WIFI	1	\$27.50
Arduino Uno R4	1	\$19.99
¾" Wood	1	\$9.99
Connectors	12	\$15.99
Total		\$131.41

Table 1. Project Parts and Prices

8.0 PROJECT IMPLEMENTATION

This is the point where hardware and software are executed to develop a functional, intelligent pill dispenser. This is the stage when theoretical plans, after design and planning, are transformed into a working prototype. For successful implementation, there should be a properly organized mechanism for the smooth integration of all components involved in the process. Some of the key steps included in the implementation of the project are indicated below.

1. Hardware Assembly

First, it is necessary to assemble the physical parts of the PillPro's system. The hardware used includes an Arduino Uno R4 WIFI, an RFID reader, servo motors, DC motors, a touchscreen, and medication hoppers. First, the physical assembly begins with designing physical housing for these components, designed in 3D print fashion attached to a wooden frame.

Servo motors attached to the top of hoppers are employed for the purpose of regulating the respective dispensed amount. Each of these hoppers is set up with different types of medication, while the servo motors provide accurate dosing dependent upon users' input. A DC motor feeds more pills into the dispenser mechanism in order for the system to keep supplying when the need arises. There is also a laser sensor that watches out for the number of pills inside each hopper and sends a notification via the mobile application when refills of medication are needed.

The system will incorporate an RFID reader, which authenticates the user. To ensure that proper medication is dispensed, either a patient or caregiver will identify themselves with RFID cards. The Arduino Uno R4 WIFI will be applied as a central controller, which connects all components and manages communication between hardware and a mobile application.

2. Software Development

Notably, alongside the hardware, software development is the crucial activity that will lead to the successful implementation of the PillPro's. Indeed, the system would make use of custom software, built using the Arduino Integrated Development Environment, with various libraries used for the handling of communication with RFID devices, motors, and WIFI connectivity.

Events controlled by the software include authenticating users using RFID, managing medication schedules, and operating the motorized dispensing mechanism. The code constantly listens for user input via the RFID reader and touchscreen. Provided a user is authenticated, it would perform the correct dispersal of medication at pre-programmed schedules.

The Arduino Uno R4 WIFI offers Wi-Fi connectivity to enable real-time communication between the pill dispenser and the mobile application. This is applicable in cases where there is a need to inform the caregiver that the quantity of medication is running low or that a scheduled dose has been missed by the user. It communicates directly with the app to provide the caregivers with the current information needed for the patient's adherence to the medication.

3. Testing and Validation

Key parts of an implementation phase involve testing and validation, where one can ensure the integrated parts work as desired. Both the software and hardware should undergo extensive testing to confirm that medication is properly dispensed by the dispenser on time, with notifications, as will be done by the app.

Initial tests are done in simulations in which the logic of the software is run in an idealistic environment in order to test for bugs or inconsistencies in the code. This enables the fine-tuning and perfecting of the team's dispensing algorithms, whereby the system can address user inputs correctly. When testing in simulation is complete, the real-world testing stage follows using the pill dispenser to dispense actual medication with schedules.

Key parameters of testing will include:

Accuracy: The system delivers the correct dose on time as per the patient's schedule.

Reliability: The system functions under varying conditions, such as different network situations.

Safety: The system prevents wrongdoing or omission and alerts caregivers of problems.

Authentication testing of users using the RFID system to check proper identification and authority of users and keeping unauthorized utilization stopped.

Any data collected during the testing phase is analyzed. The idea is to make sure that the system will meet all of the requirements of the project. Any issues that are picked up during testing are dealt with, and necessary adjustments are made to the hardware and software parts.

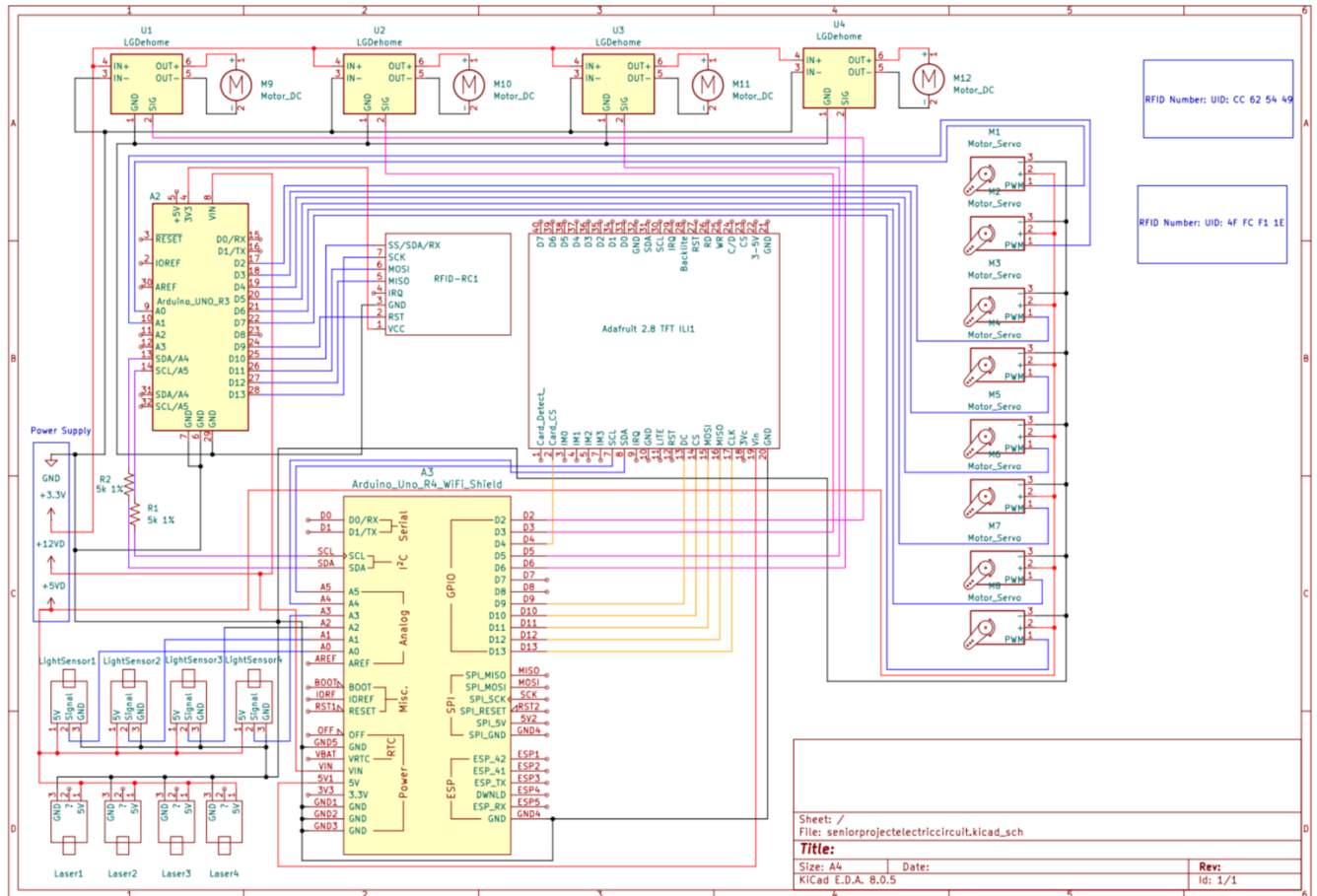


Figure 8. Wire Diagram for all system

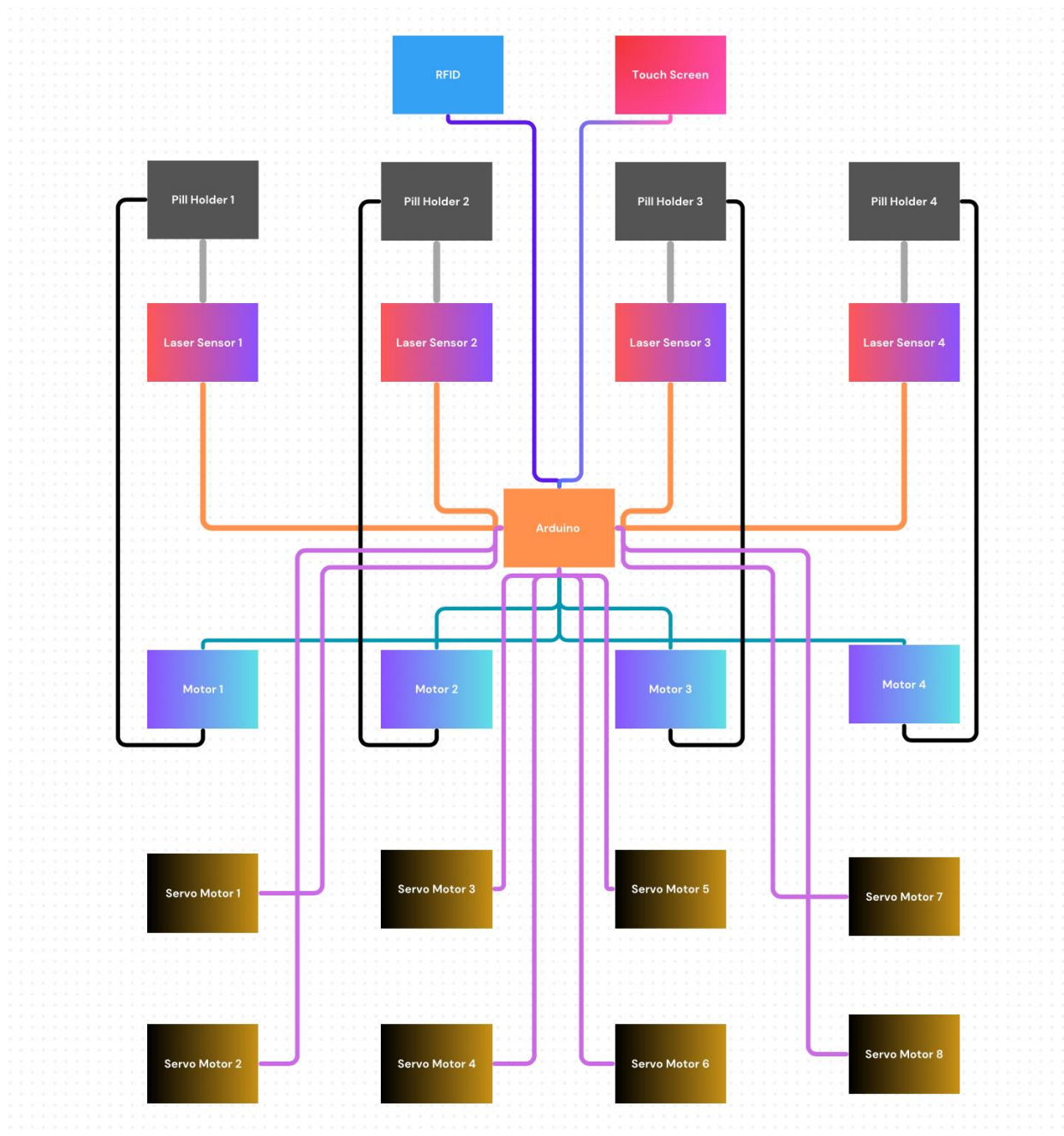


Figure 9. Block Diagram for all system

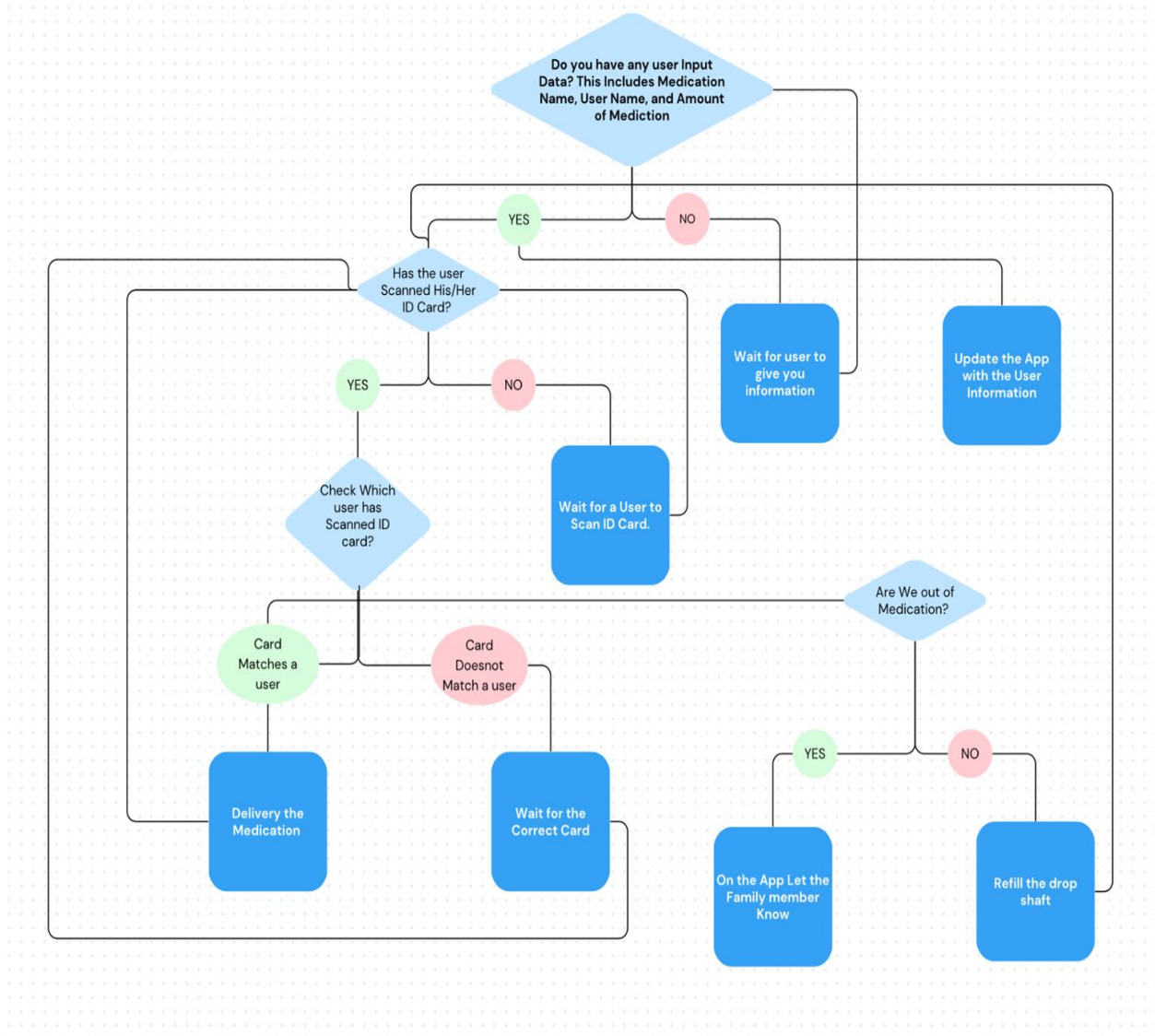


Figure 10. Programing Flow Diagram

This wiring diagram represents an intricate electronic system and was designed for use in either an advanced robotics or automation project featuring an interacting subsystem controlled by means of an Arduino-based microcontroller. Centrally lies the Arduino Uno R3, complemented with a Wi-Fi Shield acting as a processing unit which will manage inputs and outputs from a number of connected elements. The Arduino regulates the flow of data from sensors, motors, and display units in the circuit for efficient running of the entire system.

Beginning from the top of the diagram, four DC motors, M9 through M12, each connect to motor driver modules U1 to U4. These modules control the speed and direction of the motors

by shifting the polarity of the input signals, as indicated by the IN+ and IN- labels. It's assumed at this point that motors drive some mechanical elements-like wheels or actuators-powered through the 12V power supply connected via the motor driver modules.

On the right side of this diagram, there is a line of seven servo motors, M1 to M7, each driven by PWM signals. Servos are essentially utilized on applications that have something to do with precise positioning, such as controlling robotic arms or manipulating joints. These servos are linked up with the particular PWM pins on the Arduino, therefore allowing for immediate changes in positioning from the servos.

Along with the servos, the system also connects an Adafruit 2.8" TFT display-ILI1-for the visual interfacing of the user. The display would be connected with several digital pins of Arduino comprising from D0 to D13, and SPI communication pins comprising SCK, MISO, and MOSI. Important information like sensor readings, system status, and feedback from user input is likely depicted on this display, hence enhancing the usability in this project.

Lower left is a set of light sensors and laser modules. Essentially, this setup will allow the detection of changes either in the environment or in positioning. All these light sensors are connected with the Arduino's analog inputs, ranging from A0 to A5, which allows the system to process different intensities of light and react accordingly. The laser modules are for measurement purposes, alignment, or anything which would require its input being fed back into the Arduino for processing.

The power management system in the middle left of this diagram steps down an incoming 12V input to 5V and 3.3V for various component power. Voltage levels are regulated, using R1 and R2 resistors to ensure that each element receives the appropriate amount of power. Whereas the 5V powers the Arduino and most components, 3.3V would likely be reserved for low-power devices such as sensors.

Also, an RFID reader-RC1-is connected with the Arduino through SPI pins. In this context, the RFID reader should identify and read special RFID tags already included in the system for access or identification. Two numbers of RFID tags are shown in the diagram: UID: CC 62 54 49 and UID: 4F FC F1 1E, meaning these tags are previously programmed in the system.

The wiring in this diagram underlines integration of the following functions: motor control, sensing, interaction with the user through the display, and identification by means of RFID. Each subsystem is thoroughly connected to Arduino by digital and analog inputs, power, and ground lines. The diagram also shows precautions in circuit arrangements since it handles high-power devices like DC motors and low power devices, sensors, and RFID readers in an inductive design.

This may be a general circuit that corresponds to quite a complex and multifunctional project, like a robotic platform, an automated machine, or a remote-controlled system. A system designed this way would ensure that the different parts of hardware can communicate efficiently and, therefore, enable it to perform tasks related to navigation, object handling, real-time data display, while being monitored and controlled by Arduino and the Wi-Fi module.

To power the entire system, a rectifier circuit was built and implemented into a PCB. The converter will take 120VAC into a step-down transformer. This transformer steps the voltage down from 120VAC to 12VAC. This is accomplished by the transformer being a ten to one turn ratio. This transformer is also rated at ten [VA] and is UL certified. Once the voltage is through the transformer, it will be fed into the rectifier circuit. The circuit that was constructed in Multisim provided a simulation of the circuitry. The circuit and simulation results can be seen in figure 3 and figure 4 respectively. Reviewing figure 4, shows that the output of the circuit should be around 12.5VDC.

Once the simulation of the circuit was completed, the circuit could be tested on a breadboard. The breadboard will allow for the voltage measurements to be taken at the output. This is critical because the simulation provides an ideal output, whereas the physical circuit will provide the actual output results for the design. The reason for the differences in the simulation to the physical output can be various things to include slightly higher input voltage to the transformer, and different voltage drops from the rectifier diodes. These will be slightly different than the simulation and that is why the circuit will be tested on the breadboard before the design is placed into a PCB design.

We chose to use a PCB for the rectifier circuit due to multiple things. The first is that we wanted to ensure that the circuit would have the largest amount of surface area for a ground plane. The ground plane will help to provide a low-impedance path for currents to return to and

reduce the noise within the circuit. The second reason we wanted to implement the use of a PCB is for the experience of designing a PCB. This, being a common task within multiple fields, will provide a good introduction to PCB design. The third and final reason for the PCB is to provide a cleaner looking circuit for the rectifier.

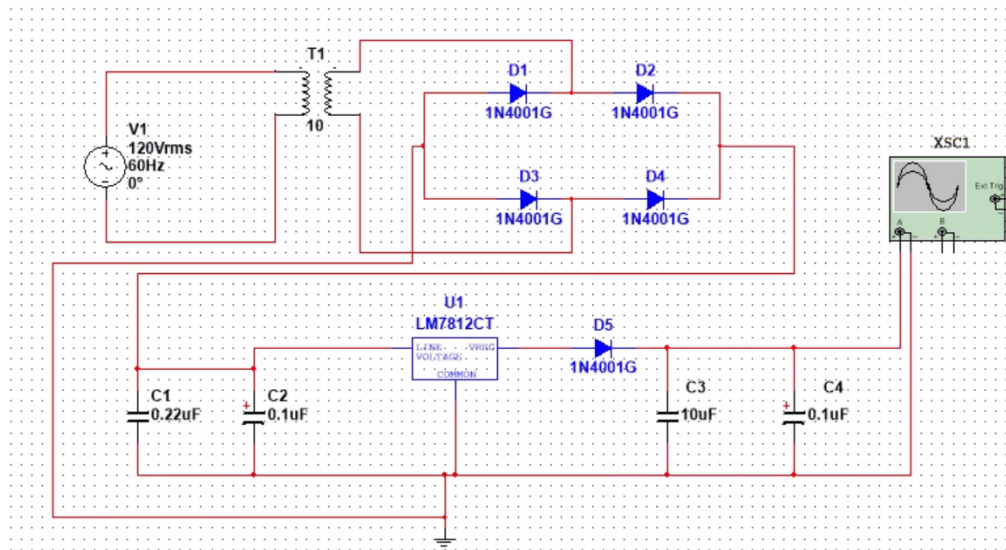


Figure 11. Rectifier Circuit in Multisim.

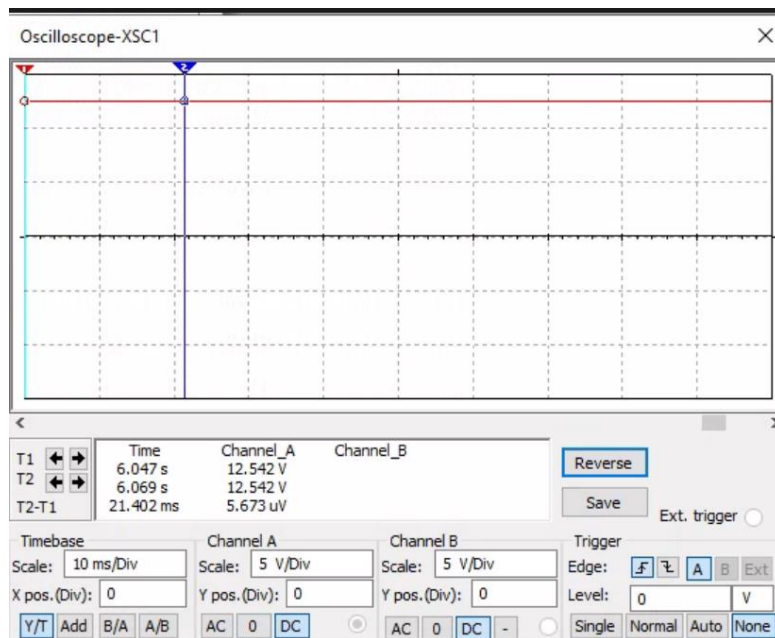


Figure 12. Oscilloscope Measurement of Rectification Circuit

9.0 PROFESSIONAL COMPONENTS

9.1 Economic Concerns

While the PillPro's project presents a bright future in the field of better medication handling, there are a few economic issues to consider, for both viability and success. The most important is the initial development cost, which may vary extremely. The hardware components required alone, like RFID readers, touchscreens, motors, and medication hoppers, are an investment in themselves. Also, software development will require adding skilled developers or even investing in solutions developed from third-party software to suit the system's needs. Yet another cost driver is prototyping, since it involves refinement of a test model of the system-characteristically not well predictable, with costs spiraling as the iterative cycle continues to meet functional requirements and improve user experience.

In regard to operational costs, there will be ongoing expenses that can be attached to the overall budget: the maintenance and repair of hardware for effective operation, periodic software updates. The consumption of power will also be a critical aspect, especially if the system requires a constant power source. The cost of electricity to operate the system, particularly if the installation includes battery backups to ensure reliability, must be added to the operating budget. Inventory management of medication hoppers will also be more expensive because of how often refills are required. Expiration of medication may result in losses that could further reduce profitability; inventory management techniques should therefore be performed to reduce waste.

Another economic concern for PillPro's are market competition. It is necessary to develop a competitive pricing strategy that can guarantee profitability by covering the costs but also be attractive for potential customers. Once similar products are available at lower prices, this might challenge the entry into the market and subsequent acceptance of PillPro's. Market demand for medication management systems might be variable, which may affect sales and revenue projections. But in such competitive dynamics, a proper strategy will make it fundamentally important to understand the target market and the unique value proposition of PillPro's.

Another big economic issue is regulatory compliance. Healthcare regulations often require considerable expenses regarding meeting their demands and acquiring various safety and

efficiency certifications. Furthermore, there is a need to provide liability insurance against any future legal problems because of the wrong medication. These are expensive and time-consuming processes, not to mention the project will need careful attention to its life cycle just to maintain such regulations.

While user training and support are essential components for any successful implementation, they are nonetheless separate costs. Training programs for users, which must be designed to enable them to work with the system efficiently, may require huge investments themselves. Furthermore, additional customer support will be required to answer user questions and resolve problems, meaning additional support personnel or a helpdesk system to enlarge the operational budget.

Lastly, the return on investment for PillPro's is critically important and needed in order to ascertain whether this will be a long-sustaining project. The stakeholders should know if their initial investment results in any type of long-term savings concerning medication management or improved health that justifies cost investment. Funding or grant opportunities may be able to be found to offset costs associated with implementing a system using this project, especially if there is possible benefit to public health. It is equally important to know the economic viability of the target market, meaning potential users are able to afford the system while getting value for their money. An economic fluctuation or downturn would further make consumers unwilling to invest in new technology. It will be of paramount importance to conduct thorough market analysis and financial planning as a strategy for weathering the uncertainty.

9.2 Environmental Concerns

In the PillPro's project, there are various concerns about the environment to be observed in the effort of making the product sustainable and as non-ecologically harmful as possible. This will relate to the material sources for the hardware components, such as plastics and metals, and the manufacturing process. Determining if these materials can be sourced sustainably is key to minimal ecological harm. In addition, the manufacturing processes for creating these components may involve several emissions and wastes, especially in companies not putting into practice environmentally friendly processes. The development of the project should give special consideration to environmental practices concerning suppliers and manufacturers.

Another important consideration is electronic waste, e-waste, from the lifecycle of components utilized in the PillPro's. Components involving touchscreens and RFID systems have rapid advancements in technology, rendering them obsolete quickly. Improperly disposing of electronic components could contaminate the environment with hazardous materials. As e-waste has become more of a problem, the development of responsible end-of-life plans is very important with such devices. This is one concern that may be mediated by encouraging users to recycle or responsibly dispose of outdated technology.

Another great environmental concern for PillPro's pertains to energy consumption. Energy use that powers the system-from the touchscreen to motors and any devices connected to it-can be a contributor to carbon emissions if the electricity applied comes from an energy source not renewable by nature. To this effect, it becomes important that energy-efficient components and practices are observed in both the design and operation of the system. In paying more attention to the ways of reducing energy use, PillPro's will be minimizing its overall carbon footprint and acting in a manner that is consonant with the goals of sustainability.

The impact of PillPro's on health facilities brings with it a host of other environmental challenges-increased use of resources such as electricity and water impact the ecological footprint of such facilities. The use of medication packaging, especially single-use materials, could further generate large volumes of waste that may strain the local waste management systems. There is therefore a need to devise ways in which waste is minimized and resources better used by health facilities.

User behavior and adherence to sustainable practices also play an important role in environmental outcomes of the project. In case the system encourages poor disposal of un-used or expired medications, it would likely translate into environmental contamination. It is highly necessary that education of users in responsible disposal methods should be highly considered in order to protect the environment from potential harm. These will also encourage users to learn about sustainable practices such as recycling, which further includes responsible management of medication that may reduce negative impacts on the environment, fostering a culture of sustainability.

Lastly, the features of climate change needs to be inculcated into the PillPro's project. The overall carbon footprint determination of the whole system right from production to transport and actual use determines the environmental impact of the system. Emission minimization through every stage of the product's life will go a long way toward ensuring its sustainability. Secondly, it becomes important to consider how these systems will adapt to the prevailing changes in environmental conditions such as high temperatures or drastic weather changes for their sustenance over the long term.

9.3 Sustainability Concerns

In developing and implementing the project PillPro's, concerns for sustainability must be on top to directly influence the aspect of long-term viability and environmental impact, entailing public perception. First, it is necessary to take care of the sustainability of materials used to build this system. Materials should be selected with a view to giving main importance to those which are not only durable and functional but also friendly to the environment. With such plastic recycling, wood harvested in a highly sustainable manner, or biodegradable material, the ecological footprint of the project could be drastically cut. Selection of suppliers that are committed to sustainability will also ensure that the raw materials used in the project are sourced and processed in a responsible manner to minimize environmental degradation along the value chain. Beyond just sourcing the raw materials, the manufacturing processes for creating the components that make up the PillPro should be highly sustainable.

This can be achieved through the adoption of clean energy technologies and the application of efficient waste reduction and pollution control techniques that minimize the level of emission and waste products produced in the factories. PillPro's will, therefore, be minimizing its impacts on climate change and natural resource depletion by partnering with sustainable manufacturers. The management of the PillPro's components at the end of their life cycles is another essential issue to be taken into consideration regarding the sustainability of the product. Considering the rate at which the technology is shifting, the touchscreens and RFID systems applied in PillPro's will become outdated within the shortest time. This makes for a growing problem of electronic waste, or e-waste, which involves significant environmental hazards if not handled properly. In view of this concern, the PillPro's project should incorporate responsible end-of-life plans for its electronic components.

This will include collaboration with recycling programs that ensure hazardous materials are disposed of in an environmentally safe way while recovering valuable resources for reuse. Besides this, the indispensable element of sustainability would be to carry out a take-back program at the company where users can return either old or unused devices for responsible recycling, so that users can be even more environmentally responsible. Another salient point of sustainability regarding the PillPro's touches upon energy consumption. In order to function properly, the touchscreen and motors need to be supplied with power in a continuous manner.

The project should aim at energy-saving technologies in which the power consumption is low, though not at the expense of performance. Besides, one may consider other sources of renewable energy for feeding the system, such as solar panels. Furthermore, users should be educated on energy conservation practices as a way of creating awareness in light of how to minimize energy consumption while on PillPro's. From the sustainability perspective, there is also a need to consider the impact of PillPro's on health facilities. With the introduction of the system, it is likely that the resource usage in health facilities, such as water and electricity, will surge, and thus health providers must be ready with practices that reduce their entire environmental impact.

Further, the packaging of medications that are, for the most part, single-use material will contribute to a great deal of waste in health care. The development of strategies that might reduce the generation of waste includes using bulk medications and/or making use of biodegradable packaging. This will require active users, healthcare providers, and stakeholders in making sustainability in culture within the PillPro's project. The key concerns that will make the project successful will involve the promotion of responsible medication disposal, minimal waste, and sustainable practices.

Indeed, addressing these sustainability issues provides a prime opportunity for PillPro's to improve its attractiveness for the ecologically minded customer, contributing to better public health and preservation of the environment. Sustainability will be a factor at every level of the design process, from the sourcing of the materials to manufacturing, including energy use and waste management, to present a system which truly reflects values of a more sustainable future.

9.4 Manufacturability Concerns

Manufacturability concerns are very critical in the development of the PillPro's project and have been a key focus in terms of the general efficiency, cost, and success of this project. The first important manufacturability concern deals with the complexity of the design of the system. While this includes just about all the other components involved in the process-RFID technology, touchscreens, servo motors, and medication hoppers-pulling it all together creates an extremely intricate assembly process that requires tight engineering and coordination. If it's too complicated, it will involve increasing manufacturing time, therefore raising labor costs, and also increasing the chances of production errors. Simplification of the design, without compromising functionality, is needed to ensure efficient manufacturability. This may involve optimization of component placement, adopting modular designs, and using standardized parts wherever possible for ease of assembling and at lower costs.

Another very important manufacturability-related issue is sourcing and availability of materials and components. PillPro's will need a lot of electronic components, such as sensors and microcontrollers, which may face disrupted supplies, be left to the mercy of fluctuating prices, and face shortages. Reliability in the supply of quality materials is very important to avoid delays in production and ensure consistent quality. In such a scenario, relationships with multiple suppliers will be helpful, along with consideration of alternative materials that can be used. Also, a well-developed supply chain management system will be highly relevant in tracking the inventory level and hence predicting any disruption in the manufacturing process to maintain smoothness and efficiency in the production process.

Cost is another critical manufacturability variable. The overall production cost has a direct influence on the pricing strategy of PillPro's and hence its market competitiveness. If the production costs are too high, there might not be a retail price that is viable for the desired customer base. To avoid this, it would be necessary that sufficient cost analyses during design identify areas where expenses can be minimized. This may include choosing cost-effective materials, optimizing manufacturing processes, and implementing lean manufacturing principles to reduce waste and enhance efficiency. By leveraging quality with cost-effectiveness, PillPro's can enhance its viability in the market and ensure it will last.

Quality control in manufacturing is also an important concern. Since the PillPro's system involves critical functions identified with medication dispensing, defects or inconsistencies in production may create serious implications for user safety and product reliability. Each part will have to be manufactured with strict quality assurance in place to ensure that each component meets the set standards. These may include scheduled inspections, testing of components, and the assembly process validation. These high standards with minimal risk of errors could be attained by training the staff in quality control practices and through the use of automated methods in quality checks.

Lastly, one manufacturability concern of the PillPro's project is the scalability of the product. In this regard, ensuring scalability of production against increased demand for the system will be important. This means a lot of planning in terms of manufacturing capacity, labor resources, and logistical arrangements. Flexible manufacturing processes can scale up or down with changed demands, hence assuring the project of unhindered growth. It is very instructive and quite strategic to try and handle growth effectively by working with manufacturers who have already gained experience in scaling their production.

9.5 Ethical Concerns

Ethical considerations form the basis of design and operation of the project PillPro's, with much consideration to the guiding principles as stated in the IEEE Code of Ethics. Of essence in consideration is the safety, health, and welfare of the public. The PillPro's system mediates the dispensing of medication and hence affects users' health. The system should be designed and implemented with the safety of its users in mind, following best practice principles in ethical design. This would include extensive testing of the technology to prevent dosing errors, whose consequences are dire for the end users. In addition to this, the project should be focused on sustainable development wherein the environmental impact of manufacturing, usage, and waste processing is as low as possible. Another important ethical issue involves the protection of users' privacy; the system PillPro's is likely to process sensitive health information, which does have a serious need for data protection measures against unauthorized access and compliance with relevant privacy regulations.

It is also an ethical responsibility to improve public understanding about the PillPro's system's capabilities and implications. Since the project involves new technologies like RFID and intelligent systems, it is important that users and stakeholders be informed about how these things actually work and their implications in society. This would allow users to actively participate in decisions regarding their health and management of medication. In addition, community outreach through the team in PillPro's would facilitate gathering feedback and responding to concerns to build a more transparent and trusted system.

Another important ethical consideration is to avoid real or perceived conflicts of interest. A development team has to make decisions that will be in the best interest of the users and the community, not their individual personal interest. Disclosing any perceived conflicts of interest to stakeholders allows the maintenance of trust in, and credibility of, the project. In addition, ethical behavior should be extended to the treatment of suppliers and partners by refusing any form of bribery or other unorthodox means that may hurt the integrity of the project in question.

Another important aspect which the PillPro's project requires is honesty about the technical work. It is absolutely crucial that mistakes be recognized and dealt with to ensure a loss of credibility and the users' trust in the system. Realistic estimations based on data and due credit to all participants should be given. This kind of culture regarding transparency and responsibility will improve not only the quality of the project but also accord with ethical standards that encourage cooperation and respect among participants.

Maintaining technical competence is one of the most important ethics that shall guide PillPro's development. Members of the team should not accept any responsibility for tasks outside their training or expertise. In such a way, the system will be designed and put into operation by those who understand the technical details and the ethical consequences of their acts. Limitations, if any, should be fully disclosed to the stakeholders in order to manage the expectations and enable them to make decisions based on complete information.

Fair treatment of all people, respect for their dignity, is also at the heart of good practice. The PillPro's team should have a friendly environment where no discrimination or harassment, on the basis of, but not limited to, race, sex, age, or disability, is manifested. A supportive workplace

culture promotes teamwork and innovation, as well as making all team members feel valued and recognized.

One would also find the need for the PillPro's team to assist colleagues in maintaining ethical standards, including discussions of ethical concerns openly and the reporting of violations without retribution. Within such a culture of accountability and ethical behavior, the team will ensure that the project PillPro's will meet not only technical and functional criteria of performance but also the highest standard of ethics to benefit users and society as a whole.

9.6 Health and Safety Concerns

The PillPro's project is an operation that will place health and safety concerns at the forefront of its development and implementation processes. The main objective of the PillPro's system is to ensure that medications are well dispensed to the right individuals for protection of the user health. It goes without saying that any failure within the dispensing mechanism might give way to serious health hazards, including quite real possibilities of medication errors that then can create adverse reactions or ineffective treatment. As such, it would be important during the design process to implement the system with various safety features that limit the possibility of such an occurrence. This would incorporate different types of fail-safes, as well as redundancy checks, to ensure that medications are appropriately identified and then dispensed. Beyond this, there should be adequate testing and validation to ensure the accuracy and reliability of the system prior to it being released into the field.

Another major health and safety issue relates to the appropriate storage and handling mechanisms of the medications stored in the PillPro's system. These four hoppers for storing different medications should be designed such that the integrity of the drugs is maintained without contamination or deterioration. It involves temperature control, humidity, and protection from light exposure, since some medications have storage conditions that should be followed for safety and effectiveness. Additionally, labeling should be clear, with an interface that is user-friendly and can assist in the proper operation of the system while minimizing the risk of mishandling or confusion regarding medication use.

Safety for the user of the PillPro's system will include not only its functional operation but also the physical design and usability. The touchscreen interface and the mechanical elements, such as servo and DC motors, are designed in such a way that their operation will not cause injuries. For instance, moving parts should be adequately guarded against any risk of pinching or entrapment, and the touchscreen design should ensure that the use of such a system presents no particular hardship for potential users of any age or with various abilities. Besides this, users should be provided with instructions and training materials to enable them to use the system safely and efficiently.

The PillPro's project also raises concerns about user privacy and data security, as sensitive health information is going to be handled. Ensuring that the user data cannot be accessed by people unauthorized is crucial not only for privacy regulation compliance but also for ensuring the trust of users in this application. Strong cybersecurity measures-for instance, encryption and secure authentication-will prove safety to personal information. The requirement for a statement of how the information to be provided by the user would be used and kept introduces some aspects of transparency and gives users grounds on which they can decide whether or not to subscribe to the system.

Besides this, there is the even greater implication for public health and safety that should be considered in the project PillPro's. The system design must include various user needs, including elderly individuals or those with disabilities that might require special assistance in managing medication. Ensuring access by all to the PillPro's system means the project can leave a positive imprint on public health and reduce health disparities among the populace.

Finally, continuous monitoring and evaluation with the introduction of the PillPro's system are important in terms of any emerging health and safety concerns. This will be accomplished through user and healthcare professional feedback, as well as examination of the usage data in hope of uncovering patterns that may lead to any potential risks. Taking necessary measures for the reporting and responding processes regarding health and safety matters should be a matter of utmost seriousness to ensure that the system remains secure and functions effectively over a period of time.

Conclusion Health and safety are the most essential perspectives to consider for bringing the PillPro's project to life. The safety of the users can be guaranteed by considering robust design, proper handling and storage of the medicine, assuring privacy to the users, and also keeping in mind a wide range of users. In this respect, public health is better off with the use of the PillPro's system while risks associated with the administration of the drug are at a minimum. Further evaluation and adaptation of the system will also enhance its reliability and effectiveness for the mission of improving the dispensing of medication both to people and healthcare providers.

9.7 Social Concerns

Social issues represent one of the cornerstones of the PillPro's project. The new medication management system is being suggested to be implemented and disseminated in various societies; thus, it is for this reason that a social issue like the question of accessibility needs to be addressed. The design of the PillPro's system must be accessible for persons of any age, ability, and technical skill. This is a system that should be very user-friendly and intuitive to ensure wide application among elderly people and people with disabilities, who might need special assistance in dealing with their medication. Large touchscreens, audio instructions, and personalized interfaces will promote accessibility within PillPro's and provide equal opportunities to all users.

The other critical social issue pertains to the impacts that the PillPro's system would have on healthcare equity. There is considerable variation in access to medication management technologies within socio-economic groups, and it will be of paramount importance to make sure that PillPro's are affordable and accessible to all. Collaborations with healthcare providers, insurance companies, and community organizations will, therefore, be of immense benefit in ensuring the PillPro's system reaches out to the neediest people in these areas. Furthermore, user education and support can be provided to allow active engagement by individuals in managing their medication and hence promotion of health equity within the community.

Another important social issue that has to be addressed in the project PillPro's pertains to privacy and data security. Because the system will handle sensitive health information, appropriate protection against unauthorized access and breaches of confidentiality should be ensured. Moreover, subjects should be properly informed about how data collection, usage, and protection are going to be carried out in order to instill confidence and transparency in the

system. Also, integrating users' privacy concerns into the design of the system will make users more confident in its use and acceptance.

Community participation is another important strategy that provides a solution to the social problems associated with PillPro's. Involvement of the community members both at the development and implementation stages of the system ensures that it meets the specific needs and preference of the population it is serving. This will be achieved by conducting surveys, focus groups, and pilot programs that outline what users might expect from the intervention and identify various reasons for resistance to or non-adoption of such an intervention. A sense of ownership given to the community members can only enhance the project's relevance to the community and thereby its effectiveness in ensuring good health outcomes.

The PillPro's project has to be seen in its larger social context of medication management technology. While the system is built to enable better individual health outcomes, it should also be considered in relation to its potential impacts on health care providers and the full health care system, including any changes in work processes in health care settings that could require additional training and support for health care professionals to use the system in their practices effectively. These workforce implications have to be addressed if PillPro's is to realize full benefits and also complementary to the existing health services.

Secondly, this project needs to take into consideration any adverse social impacts and do something to mitigate them. For instance, there could be overdependence on technology in the handling of medication, hence acting to disadvantage a certain group of people who may want medication handled traditionally or those who do not have access to the necessary technology. This would be a cause for concern, where PillPro's would have to balance between respect for individual preference and the promotion of benefits derived from technological advancement. A hybrid solution-like manual override or alternative methods for medication reminders-could help bridge the gap between conventional practice and innovative solutions.

Lastly, accountability and ethical responsibility regarding social concerns need to be cultivated in the PillPro's project. This involves a commitment not only to ensuring that the system is safe and effective but also to ongoing evaluation and improvement in light of user feedback and changes in society. With its close attention to the social implication of its

technology, PillPro's can improve not just individual community health but the social well-being of persons generally, with benefits broadly and equitably distributed.

In other words, the responsiveness to social issues is at the core of implementing the PillPro's project effectively. Emphasizing access, equity, privacy, community involvement, and ethical responsibility will facilitate the creation of such a system that can meet the diversified needs of its users and contribute towards better health outcomes and a more equitable healthcare environment. By smart design and actual community engagement, PillPro's may have the ability to make a difference in the field of medication management as well as public health.

9.8 Legal or Political Concerns

Legal and political issues hold a very important position in the realm of the development and implementation process for the PillPro's project, especially where it crosses over into the regulation of health care and personal information privacy. Major legal concerns include compliance with healthcare regulations that are required by the FDA and the Health Insurance Portability and Accountability Act of 1996. Because PillPro's controls the dispensing of medication directly, it needs to be certified to meet all regulatory standards for safety, efficiency, and user privacy. This would involve thoroughgoing testing and validation processes that would show that the technology is sound and does not present health risks to the user. A high level of attention concerning compliance with HIPAA will be expected within the system for handling sensitive patient data; therefore, protocols guarding user privacy and confidentiality are of utmost importance. Breach of such regulations may lead to substantial legal liabilities in fines, lawsuits, or both, which could put the viability of the project in serious jeopardy.

Other important legal considerations involve liability issues related to the use of the PillPro's system. The second major threat associated with this deal is if any malfunction in the system results in some error, for example-if a wrong type of medicine or medicine quantity is prescribed through malfunction resulting in clinical harm, then the big question of liability will follow on who is liable for such medication error-the manufacturer, the developer, or the healthcare professional. These are the risks associated with setting agreements and guidelines on accountability at the right place amongst the parties concerned with the project. Thirdly, the adoption of comprehensive user agreements, which spell out the limits of responsibility by the

PillPro's system, can help manage expectations and, at the same time, afford protection against potential litigation attacks.

Aside from these factors, political issues are also one of the major determinants in setting the landscape for the said PillPro's project. Healthcare policy and regulation take on an evolutionary dimension, with the rule changes placing both developers and users under uncertainty. It is clear that the cost or accessibility of the system would be tied to whatever change in government policy related to healthcare reform or technology funding within healthcare occurs. Consulting policymakers and advocacy groups can also help in making sure the project covers the current political priority or concerns of the community. Further, good relations with stakeholders in the healthcare system-not just healthcare providers, but also insurance-can ease the integration process and acceptance of the PillPro's system within prevailing healthcare frameworks.

Moreover, the PillPro's project has to address complex issues of intellectual property rights while developing novel technology for medication dispensing. Protection of Intellectual property enables the protection of innovations developed during the project, thereby enabling the company to realize its investments. It involves applying for patents on unique elements of the technology and ensuring that partnership or collaboration agreements will stipulate binding agreements relating to IP rights and its usage. If there is a failure to protect intellectual property adequately, infringement issues may arise and give rise to financial nonviability of the project.

Second, because the system PillPro's likely will be implemented in different states or countries with their different regulations, it is possible that there could be variations in respect to regulation. It may well happen that each state or country will have its own law as related to the management of medication technologies; hence, the project team has to comply with varied regulations. This complication may give rise to various scaling issues when the project is implemented in different markets. This means taking the bull by the horns in understanding and attending to regional legal requirements so that PillPro's can be legally and effectively deployed across various jurisdictions.

Finally, public perception of and confidence in the PillPro's system would also be influenced by the legal and political environment surrounding health care technologies: there is growing skepticism about data privacy, security breaches, and the misuse of personal health information.

With these concerns, the project PillPro's should be transparent and communicative with the users regarding how their data will be handled and protected. Sometimes, educational campaigns among the community help the people to be well informed and dissipate unfounded fears regarding legal and politicized issues. This will allow better acceptance and success of the project.

In conclusion, legal and political issues are very important for the successful implementation of the PillPro's project. It will involve demonstrating that the health regulations are complied with; clear liability outlines must be defined, participate in the making of policy, ensure intellectual property is protected, and determine regional differences in its legality. These will go a long way in reducing risk and further establishing viability. Furthermore, public trust will need to be achieved through transparency and community involvement, which is key to success and brings about changes in the impact on medication management and healthcare delivery.

10.0 STANDARDS

Considerations regarding guidelines and regulatory standards during design, development, and testing would ensure the quality, safety, and usability of the medication dispenser. Such standards span many key areas: quality management of medical devices, software reliability, electrical safety, data security, and usability. All combined into one comprehensive framework that ensures the medication dispenser complies with regulatory requirements and can meet user needs effectively.

The standards provide the fundamental design process, especially for those devices that have direct contact with health. ISO 13485 has been implemented in order to have a quality management system concerned with the development and production of medical devices; this ensures the dispenser follows stringent quality control right from the initial design through manufacturing to testing. This standard is quite important in ensuring that regulatory requirements concerning the production of medical devices are met. It's in addition to IEC 60601, which sets the safety and performance requirements on electrical medical equipment where contact with a patient exists. This makes sure the dispenser's electrical parts are safe and reliable to avoid any chance of electrical shock or mechanical failure. Apart from those mentioned above, ISO 14971 deals with risk management throughout the life cycle of a product. These standards guide identification of possible hazards, evaluation of risks, and measures to ensure users and caregivers are safe. The device follows safety and quality requirements as stipulated for medical applications by the preceding standards.

Software standards are equally vital for the performance and durability of the embedded system of a medication dispenser. IEC 62304 established a life cycle for the development of software in medical devices to cover design, implementation, and maintenance. Compliance with IEC 62304 ensures that the software inside the dispenser is reliable and maintainable for long-term operation in a safe way. The ISO/IEC 12207 further underpins the software development by defining best practices in development, testing, and maintenance supportive of structuring the codebase in a way capable of offering efficiency during troubleshooting and updates of the project as it evolves. These standards help evade software-related malfunctions and ensure that the dispenser runs with no glitches for a longer period to deliver seamless medication management for users.

Electronics and safety standards are proposed to hedge off risks associated with electrical failure and environmental hazards. UL 60950/UL 62368 norms for information technology and electronic equipment guarantee that the electrical parts inside the dispenser will be correctly designed for safety. This will involve controls to prevent overheating, electrical shorts, and other potential hazards. Equally important is compliance with RoHS to ensure the components of the device are free from hazardous materials such as lead or mercury in order to comply with the environmental and health standards. This combination of standards can enable the dispenser to meet safety expectations both in household and clinical settings.

Data security and privacy are also important considerations, especially if the dispenser is to record or transmit any patient information. While sensitive health data is not managed in this project at this time, compliance with HIPAA would be required for any update that might include personal health information. In the same vein, compliance with the GDPR would be required in the European market, where data protection for patient privacy is strict. Adhering to such standards within the design process would allow flexibility if, in the future, new demands in terms of secure data management arise.

Lastly, usability standards ensure that the medication dispenser will be convenient and safe for users, especially for elderly patients or those with cognitive impairments. The design principles of ISO 9241-210 focus on the use of a product by considering ease of use and accessibility. Similarly, IEC 62366 considers the usability of medical devices, which ensures that dispensers can be safely and efficiently operated by intended users. In healthcare, it is very critical that usability be given front-row attention because ease of use can be the factor that will make patients comply with their prescriptions or help improve overall health outcomes.

From quality management and software to safety, data security, and even usability, we have implemented various standards on this project. This has allowed the development of a robust framework that will support not only the immediate functionality but also the long-term reliability of the medication dispenser. The device will be functional, safe, and accessible, and also well-placed for future regulatory requirements.

11.0 MOBILE APP

The PillPro app is a novel companion tool that will realize the full functionality and usability of the PillPro device. Convenient, secure, and empowering, the app offers a seamless interface whereby users can manage their medication routines from their smartphones or tablets with ease. Its intuitive design ensures accessibility for users of all technical abilities, offering a streamlined solution to medication management challenges.

At the heart of the PillPro app is its ability to personalize medication schedules. Users can easily enter their prescriptions, including the type of medication, dosage, and timing for each dose. This information is then directly integrated into the PillPro device, which will dispense the correct pills from the medication hoppers at the pre-specified time. By reducing the risk of errors, the app plays a critical role in promoting accurate and safe medication usage.

Security is a chief feature of the PillPro app. The app lets only the authorized user view and change the medication schedule through its integration with the RFID authentication system built into the device. This is especially important in situations where multiple users would share the device, including families or assisted living facilities. Each user's profile is securely managed, safeguarding sensitive information while providing a tailored experience.

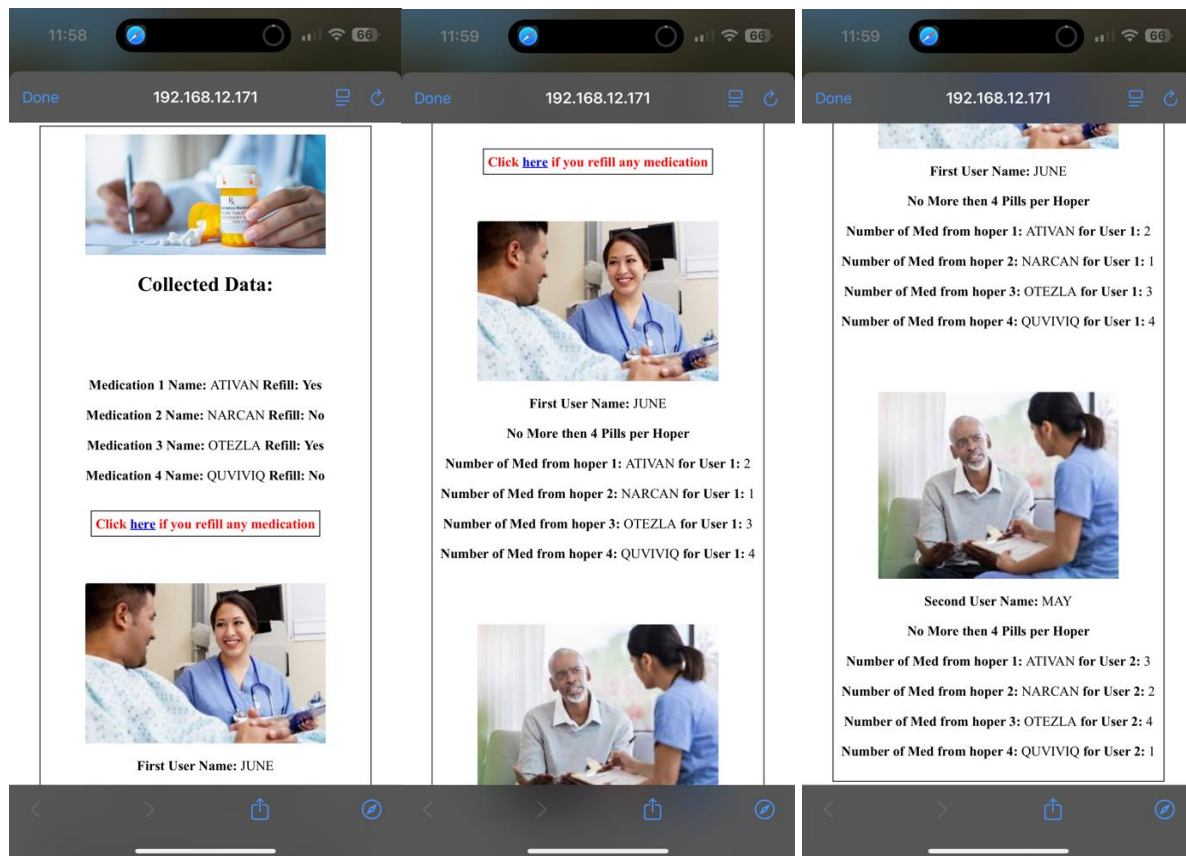


Figure 13. App layout.

Besides the scheduler and security features, the notification system integrated into the PillPro app enhances medication adherence. This sends reminders and alerts when medications are to be taken so that even the busiest life doesn't have an effect on taking proper doses on time. Users are also able to monitor, in real time, the distribution of their pills, being able to see whether the dose has been taken as ordered.

The app also contains a detailed history feature, which charts medication use over time. This can be invaluable for users who need to share adherence reports with healthcare providers or caregivers. By providing a clear overview of past medication patterns, the app supports proactive healthcare management and fosters better communication between patients and medical professionals.

The PillPro app was made with versatility and user convenience in mind. Its compatibility with multiple devices and platforms makes it accessible anywhere and anytime for

users. The user-friendly interface provides an interactive and smooth experience, and medication management is no longer a chore but has become part of daily life.

In short, the PillPro app revolutionizes the way users interact with their PillPro device by offering a comprehensive solution for medication management. By integrating scheduling, security, notifications, and usage tracking in one platform, it truly enables users to take control of their health with confidence and ease. This app will not only enhance the functionality of the PillPro device but also show its commitment to improving the overall well-being of its users.

12.0 RESULTS

Software Testing Methodology for the Project Code.

In this project, we engaged in intensive code testing methodology in order to make the software controlling the medicine dispenser reliable, accurate, and robust. It was divided into distinct stages designed to address various levels of functionality and integration. The following sections outline the various stages used in testing the code and verifying its performance across various operational scenarios.

The first level of testing was unit testing to verify each and every function in the codebase. Each module within the code, including motor control, sensor readings, and interaction via a touchscreen, was tested for expected functionality in isolation. Unit testing allowed us to isolate problems in a given function and solve them without having to debug later on with increased complexity. This had allowed confirmation of the correctness of each function before any interaction between different system functions.

Following the successful unit tests, we conducted integration testing. This level gives us the possibility to test how different code modules interact with each other to work within the system as a whole. For example, we tested how well the servo motor-controlling code is coordinated with the laser sensor for them to work in synchronization in the process of pill dispensing. Integration testing resolves all compatibility problems between components, thus introducing a smooth workflow within the medication-dispensing system.

Simulation played an important role in the testing process, as it provided a safe platform where the behavior of the code could be tested without having any effects on hardware. We utilized the use of tools such as Multisim in concert with Arduino's Serial Monitor to simulate output values and track real-time interactions in code. Such a method allowed for the identification of potential errors within a controlled environment, hence reducing the possibility of hardware damage during physical testing. These simulated values allowed for quick debugging of the code to test for preliminary accuracy standards before moving on to hardware-based testing.

For real-to-life reliability, the code was executed on the project's hardware setup in an actual Hardware-in-the-Loop test. The aim was to test how the code interacted with physical

components and the way the software was able to control hardware in operationally realistically replicated conditions. In this regard, HIL testing has been crucial to obtaining a deeper understanding of the timing, sensor responsiveness, and overall reliability of the system on real hardware, thereby optimizing the performance even further.

Stress and long-term testing focused on the robustness and stability of the system in the context of continuous use over a longer period. The code was continuously executed for long periods of time in search of memory usage, consistency in timing, and interaction of various components when the operation was sustained. Possible problems, such as memory leaks or timing drifts, have been identified and corrected at this stage to guarantee the long-term reliability of the system. Stress testing was really necessary to find those vulnerabilities that may appear after a very long period of usage and helped build resilience in the software solution.

Lastly, user acceptance testing was done during this stage, where we simulated the typical interactions by end-users with the system. The aim of this phase was to confirm that the code would meet all the usability standards related to timely performance, accuracy, and interaction with users. Real-life medication dispensing and scheduling adjustment scenarios were tested on potential users or their caregivers to make the system intuitive and reliable. Results from this phase proved very informative and provided useful feedback for the last refinements of software, increasing user-friendliness and the performance of the system as a whole.

A multi-phase testing strategy was utilized that comprehensively validated functionality, accuracy, and resiliency under countless conditions. From unit testing through user acceptance testing, each stage contributed to an overall extensively vetted software solution that could support the complex medication management requirements of the device. Ensuring this performance would be carried out reliably, safely, and effectively, this approach complied with the high standards of quality and user satisfaction set for this project.

This project was tested in phases throughout the build process to ensure that designs worked. The first test that was performed was of the hoppers that held the medicine in them. The hopper contains an auger bit within, that is connected a motor to spin it. The motor was utilizing 3.3VDC to power it and worked properly. Pills would fall from the hopper and into the funnels without jamming in the hopper or breaking pills. We found that the rate of which pills would fall from the hopper was a little too fast and sometimes would cause a jam in the funnel. This was

because too many pills would fall into the funnel, not allowing them to fall into the shout properly. We needed to slow the motors down that were connected to the augers to prevent this. We changed the input voltage to the motors from 3.3V to 2.0V. After testing the hoppers with this voltage on the motor multiple times, no jams occurred in the funnels.

With the hoppers working the correct way, the next step was to incorporate the funnels that would hold the medicine to get dispensed. The funnels have two servos that are a certain distance apart from one another. This allows for one pill to be trapped in between the servo arms. The challenge when testing this was the angle of the arms on the servo's needed to be at a particular angle when opening and closing. The angle for the open position was relatively easy to find, but the closed angle took some trial and error. Simply because if the servo arm would turn too much then it would hit into the back of the funnel, causing the servo to draw more current. The more current was drawn trying to get the arm to move to its commanded position. After finding the correct position for each servo, we found that the arms needed to be shaved to a point on the edges to help it arm get under the pill lifting it up above it on the top servo.

On the funnel we also included a laser light sensor for the purpose of indicating when a low medication is present. This light will notify the caregiver that they need to refill the medication. We designed the sensor to allow for three pills to be left in the funnel, giving a couple more doses that could be delivered before completely empty. The issue arose from the sensor sending a message of needing a refill when a pill was covering the laser light. After some troubleshooting and inspecting the inside the funnel, we determined that the laser light was extremely bright, and with the light detector only a few millimeters away from the laser, light was leaking around the pills. That light was able to be detected from the sensor. We found a filter that could be put on the laser light that would take the large beam and shrink it into a smaller laser beam. This allowed for less light to be emitted, and allowing the sensor to correctly detect when a pill was blocking the light and when no pills were in front of it.

The next part of the testing consisted of testing the power of the entire system. We built a voltage regulator circuit, that would regulate the input of 12[VDC] down to 5[VDC]. We utilized the LM7805 voltage regulator for this implementation. We included a heat sync on the voltage regulator as well to help with the heat dissipation that would be produced. While running the machine we found that the voltage regulator was still getting extremely hot. With an average

current draw of 0.6[A] and a voltage drop of 7[V], this voltage regulator needed to dissipate around 4.2[W]. The calculation to find the watts is $P = V \cdot I$. We decided that there was too much heat to have within the case of our machine. After some thought of other options, we could implement such as running two LM7805 voltage regulators in series to split the power dissipation in half or using multiple voltage regulators to step the voltage down slowly. We figured these methods wouldn't look as clean and would add complexity to the circuit. This would add more troubleshooting time if a part failed or if the circuit didn't work.

We decided to use a boost buck converter for supplying the power to the components. The converter was able to supply 12, 5, and 3.3 [VDC] from the same board and was all the voltage outputs that we needed for our components.

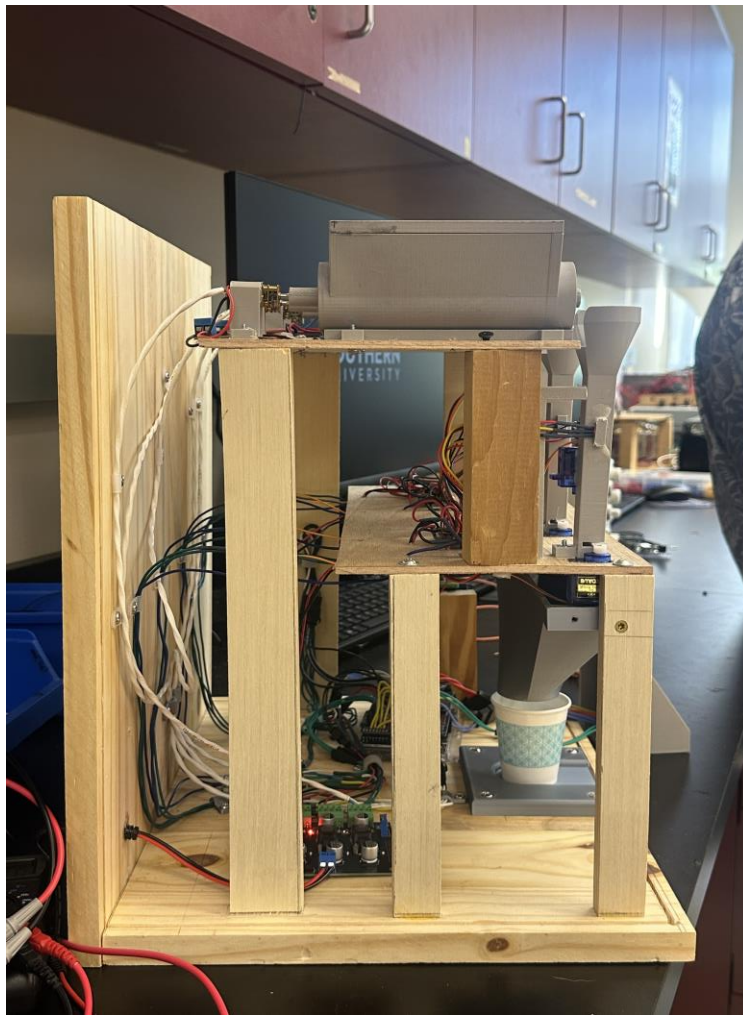


Figure 14. Side View of the project

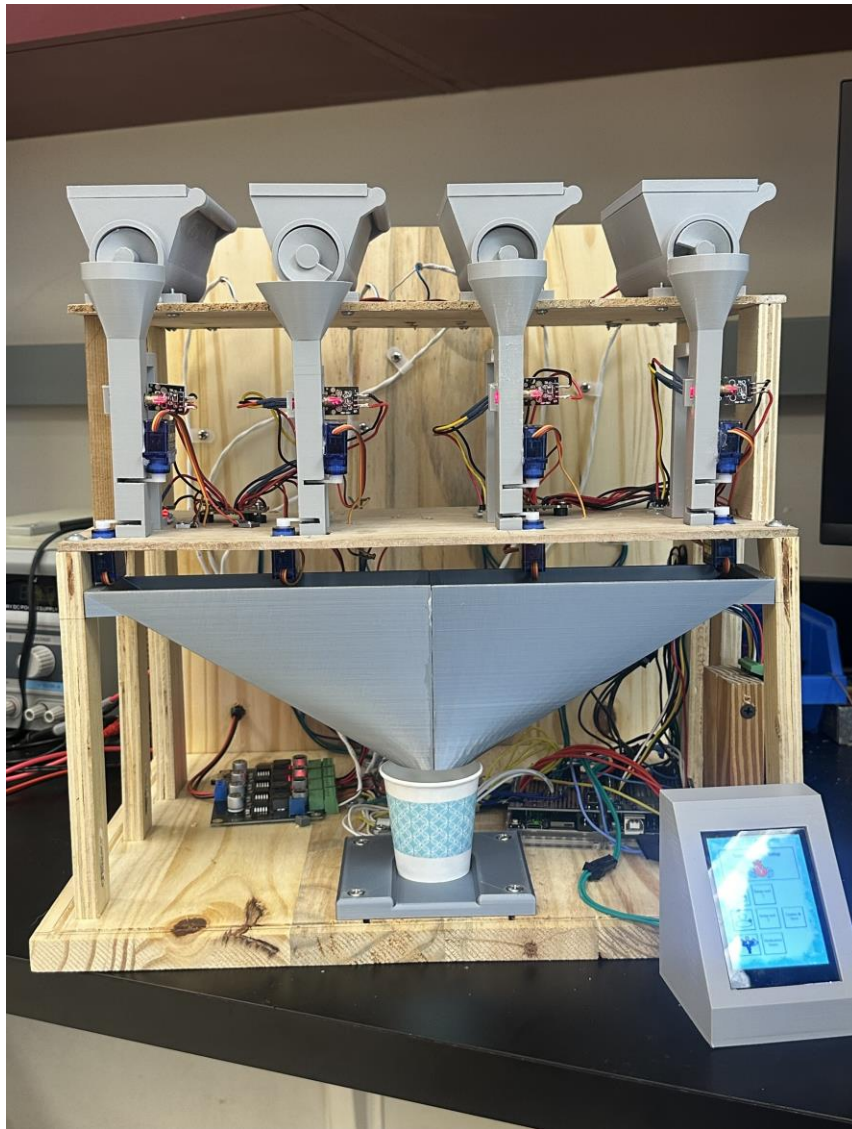


Figure 15. Front View of the Project

Test 1			
User Information	Hoper Number	Input Into Arduino	Output in the Cup
User 1: Josh	1	2	2
	2	0	0
	3	3	3
	4	4	4
User 2: Andrew	1	2	2
	2	2	2
	3	3	3
	4	4	4

Table 2. Test 1 Result of the full assembled project.

Test 2			
User Information	Hoper Number	Input Into Arduino	Output in the Cup
User 1: Josh	1	1	1
	2	3	3
	3	1	1
	4	0	0
User 2: Andrew	1	3	3
	2	0	0
	3	4	4
	4	1	1

Table 3. Test 2 Result of the full assembled project.

Test 3			
User Information	Hoper Number	Input Into Arduino	Output in the Cup
User 1: Josh	1	2	2
	2	4	4
	3	4	4
	4	4	4
User 2: Andrew	1	2	2
	2	1	1
	3	1	1
	4	4	4

Table 4. Test 3 Result of the full assembled project.

Test 4			
User Information	Hoper Number	Input Into Arduino	Output in the Cup
User 1: Josh	1	3	3
	2	2	2
	3	4	4
	4	1	1
User 2: Andrew	1	2	2
	2	2	2
	3	4	4
	4	4	4

Table 5. Test 4 Result of the full assembled project.

Test 5			
User Information	Hoper Number	Input Into Arduino	Output in the Cup
User 1: Josh	1	1	1
	2	2	2
	3	2	2
	4	3	3
User 2: Andrew	1	3	3
	2	4	4
	3	2	2
	4	3	3

Table 6. Test 5 Result of the full assembled project.

Test 6			
User Information	Hoper Number	Input Into Arduino	Output in the Cup
User 1: Josh	1	3	3
	2	1	1
	3	1	1
	4	2	2
User 2: Andrew	1	4	4
	2	2	2
	3	3	3
	4	1	1

Table 7. Test 6 Result of the full assembled project.

Test 7			
User Information	Hoper Number	Input Into Arduino	Output in the Cup
User 1: Josh	1	4	4
	2	2	2
	3	3	3
	4	3	3
User 2: Andrew	1	4	4
	2	4	4
	3	3	3
	4	1	1

Table 8. Test 7 Result of the full assembled project.

Test 8			
User Information	Hoper Number	Input Into Arduino	Output in the Cup
User 1: Josh	1	1	1
	2	2	2
	3	4	4
	4	2	2
User 2: Andrew	1	4	4
	2	1	1
	3	4	4
	4	2	2

Table 9. Test 8 Result of the full assembled project.

Test 9			
User Information	Hoper Number	Input Into Arduino	Output in the Cup
User 1: Josh	1	1	1
	2	2	2
	3	3	3
	4	2	2
User 2: Andrew	1	2	2
	2	4	4
	3	3	3
	4	3	3

Table 10. Test 9 Result of the full assembled project.

Test 10			
User Information	Hoper Number	Input Into Arduino	Output in the Cup
User 1: Josh	1	4	4
	2	2	2
	3	4	4
	4	4	4
User 2: Andrew	1	2	2
	2	3	3
	3	1	1
	4	1	1

Table 11. Test 10 Result of the full assembled project.

13.0 DELIVERABLES

The project deliverables in the medication dispenser provide the details on each phase of the design, starting from initial design to testing and final documentation to ensure that the device will be able to meet user requirements on safety. The listed deliverables are constituent elements that track, communicate progress, support testing, and ensure functional end products.

Starting with the Project Proposal and Design Specifications Document, it outlines the objectives and aims of the medication dispenser by identifying target users and details of the primary features, components, and functionalities. This document lays the framework for development by highlighting what needs to be achieved in the project and forms the basis for the technical requirements to meet the user's needs. The second major deliverables are Hardware and Software Prototypes that aim to provide proof of concept. The hardware prototype has basic casing, a touchscreen, an RFID reader, and compartments for medication. The software prototype covers the basic code and UI design. They enable early feasibility testing of core functionalities of the device.

The Wire Diagram and Circuit Schematics will be very important in understanding and managing the electronic layout. This diagram shows the interfacing of every piece, from sensors right through to motors, providing an extremely valuable reference as the project advances when troubleshooting or testing is required. Additionally, a 3D Printed Model of Mechanical Components has been prepared, ensuring all the components-like the medication hopper-will fit together as expected. This model can be used for fit testing prior to final assembly, reducing chances of mechanical errors.

The Gantt Chart of the Project Timeline is a planning tool that outlines the specific timeline to be followed for the development process, including design, the development of the prototype, testing, and final adjustments. It provides specific deadlines for each of the deliverables while enabling the team to keep on track, allowing flexibility for testing and adjustment. This finds complementation in the Parts List and BOM, which is a comprehensive document listing components with specifications, quantities, cost, and sources. The record is further important because it plays a vital role in budgeting and replication in later times, particularly for scaled-up production.

Software code has a critical role in the dispenser; hence, to make the code more understandable and thereby explain its structure in terms of control sequences and variables with comments for critical functions, the Programming Code Documentation is prepared accordingly. The provided documentation enables others to have a more understandable code, and hence easily debug and perform updates over time. To the end of the project, the Testing and Quality Assurance Report summarizes the testing methodologies and results by detailing functional, user, and reliability testing phases. The report will further flag any adjustments made based on test outcomes to make certain the device meets the required standards of safety and usability.

After testing, the details of the project, from design considerations to setbacks and solutions, as well as final results, are outlined in the Final Project Report. This gives a comprehensive overview of the project, reflecting upon success and consideration of areas for future improvement. Development of a User Manual and Maintenance Guide further supports this work by outlining use instructions for the caregivers and the end-users regarding device operations, troubleshooting, and maintenance. The guide enables the medication dispenser for real-world use effectively and safely, extending usability and value.

Each of these deliverables points to the goal of the project in developing a medication dispenser that is safe, effective, and easy to use. These papers range from initial design documents to a final project report and establish a sound framework through which development, testing, and user adoption occur.

14.0 FAILED IDEAS

During the initial planning, we had several ideas of how we could implement an automatic pill dispenser. We knew that the most challenging part of this project would be how the pills are stored in the machine and then how they are dispensed. Our initial idea was to fill a funnel with the medication, then to use servo motors to dispense the medication. We started to implement this plan by 3D printing a funnel that would hold the medication. The first design of the funnel can be seen in figure 13. below.



Figure 16. Initial prototype for pill holder

Once the funnel was printed, we began testing by placing medication into the funnel and checking to see if the medicine would drop out one at a time. This is where we came to the first design flaw of the project. The funnel would work for round pills, but when it came to elongated pills, they would get stuck dropping into the smaller piece of the funnel. We had an idea to use a rotating mass vibrating motor to help shake around the pills in the funnel to prevent any of them getting stuck. This would have worked we believe, but when it came to mounting the vibrating

motor to platform that the funnel would sit on, we saw that it would end up shaking the entire case of the machine. Over time with this design, we believed that this could cause other components to fail prematurely or cause the casing to shake when placed on a countertop.

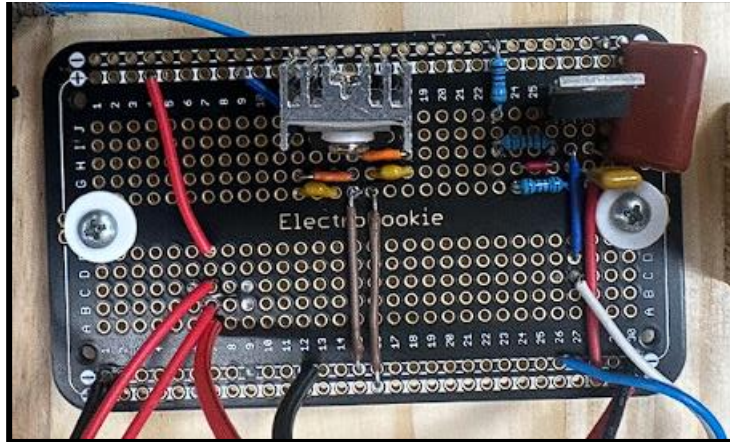
This led to us needing to scrap this design and go back to the drawing board. We came up with another idea to use that would hold the medicine and be able to dispense the pills one at a time. The plan was to use an auger style bit that would sit inside of a casing with a funnel on top of it. The funnel would have a larger opening to allow for the pills to fall into the auger without getting jammed from one another. The design can be seen in figure 14. This design was promising as it could dispense the medication with the use of a motor to turn to the auger bit.



Figure 17. Second prototyped idea

This idea was promising and worked for feeding the pills one at a time, but an occasional blockage would occur where the pills would drop down into the auger bit. The pills would then get stuck in the opening, not allowing the auger to turn anymore. The basic concept was there and with a little tweaking of the funnel on top, we would then come to our final design of how the medication would be stored in the machine and be able to be dispensed. This design would lead to the final design that would be medicine holder for different shapes and sizes of medication.

Another significant challenge we faced during development was with the power supply. The system required a voltage converter circuit to step down 12 volts DC to 3.3 volts for some of the components. For this, we used an LM7805 voltage regulator. The design can be seen in figure 15.



However, we quickly realized that the circuit was producing too much heat. At one point, we noticed a burning smell, indicating that the regulator was overheating. This overheating likely caused the LM7805 to hit its maximum operating temperature and enter thermal shutdown mode. The LM7805 is designed with thermal protection, meaning that when it gets too hot (around 125°C to 150°C), it automatically shuts down to prevent damage, cutting off power to the circuit.

Once it cooled down, the regulator resumed normal operation, but the recurring heat issue highlighted the need for better thermal management. Prolonged overheating could lead to component degradation over time, reducing reliability. The high heat was likely due to the large voltage drop from 12 volts to 3.3 volts, combined with the current demands of the system. This forced us to rethink our power regulation strategy and consider alternative solutions, such as switching to a more efficient design to prevent thermal cycling and ensure system stability.

15.0 DISCUSSION

The PillPro's project represents a major step forward in the field of technology related to medication management, in an effort to increase the accuracy and efficiency of the distribution of medication to patients, thereby eliminating most-if not all-of the many concerns associated with its use. The interaction between ethical, health and safety, social, legal, and political considerations underline the difficulty of developing a system that would be able not only to match the expectations set forth by the technologies involved but also to meet diverse user needs and comply with regulatory requirements.

Central to the PillPro's project is the commitment to uphold ethical standards in safeguarding user health and privacy. Following the IEEE Code of Ethics, the projects underline the importance of integrity and responsible behavior during design and implementation. Ensuring that the system considers users' safety is not only a legal requirement but an ethical one; medication errors can result in life-altering consequences. This commitment to safety is inextricably entwined with health and safety concerns; hence, a strong test and validation process will be required to be assured that the project minimizes risks associated with medication dispensing. The requirement for tight quality control and ease-of-use interfaces further reflects the commitment of the project to a safe, accessible environment for all users, but in particular the vulnerable elderly or those with various disabilities.

Social issues also form part of the successful development of the PillPro's system. Basic access and equity will determine whether all persons will be able to benefit from the capability of the project despite their status in terms of socioeconomic and technological literacy. During the development phase, community engagement develops a sense of ownership and relevance; consequently, the public health outcomes increase. Besides that, privacy handling and data security require not only compliance with the law but also trust-building within the community. These are social issues, one leading to the other, which indicate how such user involvement, from concept to through the life of the project, demands transparency.

Legal and political considerations further complicate the landscape for PillPro's. The legitimacy and safety basis for the system remains with healthcare regulations imposed, for example, by the FDA and HIPAA. However, navigating the issues of liability, intellectual

property rights, and regional legal variations is considerably complicated. Active steps will be taken to understand and address these legal frameworks as an important mitigant to ensure the viability of the project in various jurisdictions. Engagement on such fronts with policymakers and stakeholders at so many levels might ease integration into the health system, bring the PillPro's project in line with current political priorities, and respond to community needs.

Of course, here lies the key to success-whether the project addresses all these concerns in a comprehensive and cohesive manner or not. The interplay between ethical, health, and safety; social; legal; and political issues calls for a consultative network that involves not only stakeholders but also health caregivers, community-based organizations, and regulating authorities. It is through this open dialogue and constant assessment that the PillPro's team will find easy accommodation of emerging challenges and opportunities to keep the system relevant, effective, and helpful in securing better medication management and health outcomes.

In fact, the PillPro's project is an approach toward innovation with ethical responsibility and community involvement. The project will be able to make a much more significant contribution to the healthcare environment with the myriad issues accompanying the design and implementation of medication management technology. By building robust safety measures into our architecture, considering mechanisms of accessibility and equity, and following the legal and ethical standards within the field, PillPro's will be well-placed to play an important role both in the improvement of medication adherence and, generally, in improving patient outcomes.

16.0 CONCLUSIONS

The whole design of the medication dispenser has been developed in such a way that the growing need for effective and practical medication management-mostly required by patients with complex health regimes or cognitive impairments-would be served in a realistic, reliable, and user-friendly manner. Within the scope of this project, our approach has been meticulous, following rigorous standards for design, testing, and user accessibility. For that reason, we have made sure that the dispenser will meet the highest possible standards concerning safety, accuracy, and ease of use by making both functionality and user experience of high priority.

Starting from a clearly defined set of objectives and user requirements, the project set out to design a device which could automate medication management. The design process was anchored in detailed research and strategic planning, which allowed us to address key technical and functional requirements very early in the project. As a result of the clear specifications outlined within the Project Proposal and Design Specifications Document, we were able to keep this project aligned with our goals and make sure that each phase of development built on solid ground. This set out critical parameters, target users, and initial design concepts, allowing us to refine the core functionality of the device.

Successive iteration through hardware and software prototypes was used to validate the feasibility of our design while incorporating adjustments based on real-time testing feedback. The prototypes included a basic casing, touchscreen interface, RFID reader, medication compartments, initial software to navigate through these functionalities, and. We were thus in a position to find out and correct well in advance, various problems that could have proved to be a disaster later on, concerning the fit and interaction of components, which meant that everything worked perfectly. Here again, the Wire Diagram and Circuit Schematics showed the electronic layout and supported troubleshooting and maintenance because they detailed how all the components interact with one another. Of importance to the project was the development of a 3D Printed Model of Mechanical Components.

This model allowed us to physically test the location and setup of the device's most important components, such as the medication hopper, for compatibility before actual assembly was performed. The Gantt Chart for Project Timeline kept the team on the right track concerning the deadlines for design, prototyping, and testing phases of the projects, while at the same time

giving room for flexibility if unforeseen challenges were to arise. This is further made cost-effective and reproducible with the Parts List and Bill of Materials, which specified every component, its quantity, and source. This forms a very important reference for both current and future project work. The dispenser was centrally programmed, and detailed documentation was done through Programming Code Documentation. We annotated the code and explained why sequences of control, functions, and key variables were used; hence, this forms a trustworthy document that will be used for debugging and future improvements. This documentation of the code is essential for the person maintaining or developing the project further because it explains the logic behind every line of code, to which it further contributes for the longevity of the device.

As the testing and quality assurance phases progressed, this Testing and Quality Assurance Report documented the process of testing, methodology, and results of the same.

This report provides all detailed steps necessary to ensure that the dispenser is safe, reliable, and functions as intended. We are functionally testing, user testing, and reliability-checking to verify that the dispenser performs as intended in real life by making any adjustments necessary prior to deployment. This was an important phase of rigorous testing necessary to ensure the dispenser was safe and easy for both patients and caregivers to use. The culmination of this project is represented by the Final Project Report, synthesizing all phases of development and pointing out both accomplishments and areas where potential improvement might be addressed.

The following report details the design process, challenges experienced, solutions adopted, and final results, which can give useful insight to help in further advancements. In addition, a User Manual and Maintenance Guide will be prepared to ensure that users operate the dispenser with much ease. The paper contains simple methods through which users of all levels of technical skill can put the dispenser into service, operate it, and service any potential issues. The maintenance manual further extends the device's life with good practices for maintenance and component replacement. Bringing it all together, from design specifications to the final report, the deliverables have all come together to provide the umbrella goal of the project: a reliable, efficient, and user-friendly medication dispenser.

All careful planning, testing, and documentation resulted in a device that not only met functional requirements but provided an intuitive user experience. We will provide such a solution with this final product, which has the potential of promoting compliance by reducing the risk of missed doses and improving the quality of life of patients and caregivers. By solving the critical problems in medication management, this project has established this treatment dispenser as one of the promising devices in the healthcare system for even wider research areas for improvement and application in the future.

17.0 REFERENCES

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APPENDICES

APPENDIX A. Arduino Code for Project Leader code:

```
#include <Adafruit_GFX.h>      // Core graphics library
#include <Adafruit_ILI9341.h>   // Hardware-specific library
#include <SdFat.h>              // SD card & FAT filesystem library
#include <Adafruit_ImageReader.h> // Image-reading functions
#include <Adafruit_FT6206.h>    // Touchscreen library
#include "arduino_secrets.h"
#include "WiFiS3.h"

#define KEY_X 10
#define KEY_Y 60
#define KEY_W 40
#define KEY_H 28
#define KEY_MARGIN 5
#define SPACE_KEY_W 50
#define RETURN_KEY_W 50
#define KEYBOARD_WIDTH 5
#define KEYBOARD_HEIGHT 8
#define SD_CS 4 // SD card select pin
#define TFT_CS 10 // TFT select pin
#define TFT_DC 9 // TFT display/command pin
#define TS_CS 6 // Touchscreen select pin
#define DETECT1 A0 // pin A0 for sensor
#define DETECT2 A1 // pin A1 for sensor
#define DETECT3 A2 // pin A2 for sensor
#define DETECT4 A3 // pin A3 for sensor
#define ACTION1 2 // pin 2 for action to do something
#define ACTION2 3 // pin 3 for action to do something
#define ACTION3 5 // pin 5 for action to do something
#define ACTION4 6 // pin 6 for action to do something

// Constants
const unsigned long timeout = 60000; // 1 minute timeout in milliseconds

// Variables to store start times
unsigned long startTime1 = 0;
unsigned long startTime2 = 0;
unsigned long startTime3 = 0;
unsigned long startTime4 = 0;

// Flags to indicate if the action is waiting for one false and one true detection
bool action1On = false;
bool action2On = false;
bool action3On = false;
bool action4On = false;
bool waitForReset1 = false;
```



```

bool waitForReset2 = false;
bool waitForReset3 = false;
bool waitForReset4 = false;
bool skipSensor1 = false;
bool skipSensor2 = false;
bool skipSensor3 = false;
bool skipSensor4 = false;

String Refill1 = "";
String Refill2 = "";
String Refill3 = "";
String Refill4 = "";
String Med1Name = "";
String Med2Name = "";
String Med3Name = "";
String Med4Name = "";
String User1Name = "";
String User2Name = "";
String userInput;

int Number2ofMed1;
int Number2ofMed2;
int Number2ofMed3;
int Number2ofMed4;
int Number1ofMed1;
int Number1ofMed2;
int Number1ofMed3;
int Number1ofMed4;
//int NumberTime1;
//int NumberTime2;

char keys[KEYBOARD_HEIGHT][KEYBOARD_WIDTH] = {
    {'0','1','2','3','4'},
    {'5','6','7','8','9'},
    {'Q','W','E','R','T'},
    {'Y','U','I','O','P'},
    {'A','S','D','F','G'},
    {'H','J','K','L','Z'},
    {'X','C','V','B','N'},
    {'M',' ',' ','<','-'}};

String inputString = ""; // a String to hold characters
unsigned long lastKeyPress = 0; // Variable to store the last key press time
unsigned long debounceDelay = 200; // Delay between key presses (milliseconds)
const int resetButtonPin = 7; // Update this to the correct pin number
char ssid[] = SECRET_SSID; // your network SSID (name)
char pass[] = SECRET_PASS; // your network password (use for WPA, or use as key for WEP)
int keyIndex = 0; // your network key index number (needed only for WEP)

```

```

int led = LED_BUILTIN;
int status = WL_IDLE_STATUS;
WiFiServer server(80);

// Function prototypes
void check();
String handleAction(int detected, int actionPin, unsigned long &startTime, bool &actionOn, bool &waitForReset);

Adafruit_ILI9341 tft = Adafruit_ILI9341(TFT_CS, TFT_DC);
Adafruit_FT6206 ts; // Declare touchscreen object
SdFat SD; // SD card filesystem
Adafruit_ImageReader reader(SD); // Image-reader object, pass in SD filesystem

void setup(void) {
  Serial.begin(115200);
  pinMode(resetButtonPin, INPUT_PULLUP);
  pinMode(led, OUTPUT); // set the LED pin mode

  pinMode(DETECT1, INPUT); //define detect input pin
  pinMode(DETECT2, INPUT); //define detect input pin
  pinMode(DETECT3, INPUT); //define detect input pin
  pinMode(DETECT4, INPUT); //define detect input pin

  pinMode(ACTION1, OUTPUT); //define ACTION output pin
  pinMode(ACTION2, OUTPUT); //define ACTION output pin
  pinMode(ACTION3, OUTPUT); //define ACTION output pin
  pinMode(ACTION4, OUTPUT); //define ACTION output pin

  tft.begin();
  ts.begin(TS_CS);

  Serial.print(F("Initializing filesystem..."));
  if(!SD.begin(SD_CS, SD_SCK_MHZ(25))) {
    Serial.println(F("SD begin() failed"));
    for(;;);
  }
  Serial.println(F("OK!"));

  tft.fillScreen(ILI9341_BLACK);

  Serial.print(F("Loading Connect.bmp to screen..."));
  loadImage("/Connect.bmp");

  // check for the WiFi module:
  if (WiFi.status() == WL_NO_MODULE) {
    Serial.println("Communication with WiFi module failed!");
    // don't continue

```

```

    while (true);
}

String fv = WiFi.firmwareVersion();
if (fv < WIFI_FIRMWARE_LATEST_VERSION) {
    Serial.println("Please upgrade the firmware");
}

// attempt to connect to WiFi network:
while (status != WL_CONNECTED) {
    Serial.print("Attempting to connect to Network named: ");
    Serial.println(ssid);          // print the network name (SSID);

    // Connect to WPA/WPA2 network. Change this line if using open or WEP network:
    status = WiFi.begin(ssid, pass);
    // wait 10 seconds for connection:
    delay(10000);
}

server.begin();                  // start the web server on port 80
printWifiStatus();               // you're connected now, so print out the status
Serial.print(F("Loading LayOut.bmp to screen..."));
ImageReturnCode stat = reader.drawBMP("/LayOut.bmp", tft, 0, 0);
reader.printStatus(stat);
Wire.begin();                   // join i2c bus (address optional for master)
delay(350); // Pause 2 seconds before moving on to loop()
}

void loop() {
    // Check if medication is needed
    check();

    // Get current time

    // Check if there is a touch
    if (ts.touched()) {
        digitalWrite(ACTION1, LOW);
        digitalWrite(ACTION2, LOW);
        digitalWrite(ACTION3, LOW);
        digitalWrite(ACTION4, LOW);
        // Read touch point
        TS_Point p = ts.getPoint();
        // Scale touch coordinates to match the screen resolution
        p.x = map(p.x, 0, 240, tft.width(), 0);
    }
}

```

```

p.y = map(p.y, 0, 320, tft.height(), 0);

// Check if touch point is in the bottom left
if (p.x < 80 && p.y > 240) {
  // Load User1N.bmp
  loadImage("/Med1.bmp");
  delay(1000);
  while (!ts.touched());
  Med1Name = getUserInput();

  loadImage("/Med2.bmp");
  delay(1000);
  while (!ts.touched());
  Med2Name = getUserInput();

  loadImage("/Med3.bmp");
  delay(1000);
  while (!ts.touched());
  Med3Name = getUserInput();

  loadImage("/Med4.bmp");
  delay(1000);
  while (!ts.touched());
  Med4Name = getUserInput();

  loadImage("/MedDone.bmp");
  delay(5000);
  ImageReturnCode stat = reader.drawBMP("/LayOut.bmp", tft, 0, 0);
}

// Check if touch point is in the middle left
else if (p.x < 80 && p.y > 160 && p.y < 240) {
  // Load User2N.bmp
  loadImage("/User2N.bmp");
  delay(1000);
  while (!ts.touched());
  User2Name = getUserInput();

  loadImage("/MedSel1.bmp");
  delay(1000);
  while (!ts.touched());
  userInput = getUserInput();
  // Convert String to int
  Number2ofMed1 = userInput.toInt();
  if (Number2ofMed1 > 4){
    Number2ofMed1 = 0;
  }

  loadImage("/MedSel2.bmp");

```

```

        delay(1000);
        while (!ts.touched());
        userInput = getUserInput();
        Number2ofMed2 = userInput.toInt();
        if (Number2ofMed2 > 4){
            Number2ofMed2 = 0;
        }

        loadImage("/MedSel3.bmp");
        delay(1000);
        while (!ts.touched());
        userInput = getUserInput();
        Number2ofMed3 = userInput.toInt();
        if (Number2ofMed3 > 4){
            Number2ofMed3 = 0;
        }
        loadImage("/MedSel4.bmp");
        delay(1000);
        while (!ts.touched());
        userInput = getUserInput();
        Number2ofMed4 = userInput.toInt();
        if (Number2ofMed4 > 4){
            Number2ofMed4 = 0;
        }
    /*
        loadImage("/Set.bmp");
        delay(1000);
        while (!ts.touched());
        userInput = getUserInput();
        NumberTime2 = userInput.toInt();
    */
    loadImage("/U2Done.bmp");
    delay(5000);
    ImageReturnCode stat = reader.drawBMP("/LayOut.bmp", tft, 0, 0);
}

// Check if touch point is in the top left
else if (p.x < 80 && p.y < 160) {
    // Load Med1.bmp
    loadImage("/User1N.bmp");
    delay(1000);
    while (!ts.touched());
    User1Name = getUserInput();

    loadImage("/MedSel1.bmp");
    delay(1000);
    while (!ts.touched());
    userInput = getUserInput();
    Number1ofMed1 = userInput.toInt();

```

```

    if (Number1ofMed1 > 4){
        Number1ofMed1 = 0;
    }

    loadImage("/MedSel2.bmp");
    delay(1000);
    while (!ts.touched());
    userInput = getUserInput();
    Number1ofMed2 = userInput.toInt();
    if (Number1ofMed2 > 4){
        Number1ofMed2 = 0;
    }

    loadImage("/MedSel3.bmp");
    delay(1000);
    while (!ts.touched());
    userInput = getUserInput();
    Number1ofMed3 = userInput.toInt();
    if (Number1ofMed3 > 4){
        Number1ofMed3 = 0;
    }

    loadImage("/MedSel4.bmp");
    delay(1000);
    while (!ts.touched());
    userInput = getUserInput();
    Number1ofMed4 = userInput.toInt();
    if (Number1ofMed4 > 4){
        Number1ofMed4 = 0;
    }
}

/*
    loadImage("/Set.bmp");
    delay(1000);
    while (!ts.touched());
    userInput = getUserInput();
    NumberTime1 = userInput.toInt();
*/

loadImage("/U1Done.bmp");
delay(5000);
ImageReturnCode stat = reader.drawBMP("/LayOut.bmp", tft, 0, 0);
}

else if (p.x > 80 && p.x > 160 && p.y < 260) {
    // Display collected string data
    tft.fillScreen(ILI9341_BLACK);
    tft.setCursor(0, 0);
    tft.setTextSize(2);

    // Display collected data from User1

```

```

tft.println("User1 Meds: " + User1Name);
//tft.println("Daily Doses: " + String(NumberTime1));
tft.println(Med1Name + ": " + String(Number1ofMed1));
tft.println(Med2Name + ": " + String(Number1ofMed2));
tft.println(Med3Name + ": " + String(Number1ofMed3));
tft.println(Med4Name + ": " + String(Number1ofMed4));
tft.println(" ");
// Display collected data from User2
tft.println("User2 Meds: " + User2Name);
//tft.println("Daily Doses: " + String(NumberTime2));
tft.println(Med1Name + ": " + String(Number2ofMed1));
tft.println(Med2Name + ": " + String(Number2ofMed2));
tft.println(Med3Name + ": " + String(Number2ofMed3));
tft.println(Med4Name + ": " + String(Number2ofMed4));

Wire.beginTransaction(9); // transmit to device #9
Wire.write(100);          // sends "100" to slave
Wire.endTransmission();  // stop transmitting

delay(50);

// Send data to another Arduino
sendDataToArduino();

while (!ts.touched());
ImageReturnCode stat = reader.drawBMP("/LayOut.bmp", tft, 0, 0);
}
}
delay(200); // Pause 200 ms before next touch reading
WiFiClient client = server.available(); // listen for incoming clients

if (client) { // if you get a client,
  Serial.println("new client"); // print a message out the serial port
  String currentLine = ""; // make a String to hold incoming data from the client
  while (client.connected()) { // loop while the client's connected
    if (client.available()) { // if there's bytes to read from the client,
      char c = client.read(); // read a byte, then
      Serial.write(c); // print it out to the serial monitor
      if (c == '\n') { // if the byte is a newline character

        // if the current line is blank, you got two newline characters in a row.
        // that's the end of the client HTTP request, so send a response:
        if (currentLine.length() == 0) {
          // HTTP headers always start with a response code (e.g. HTTP/1.1 200 OK)
          // and a content-type so the client knows what's coming, then a blank line:
          client.println("HTTP/1.1 200 OK");
          client.println("Content-type:text/html");
          client.println();

```

```

/*
// Print the collected data from the touchscreen with formatting
client.println("<html><head><style>body { background-color: #8fce00; background-size: cover; text-align:
center; font-size: 34px; } .data-box { border: 2px solid black; padding: 20px; display: inline-block;
}</style></head><body>");
client.println("<div class='data-box'>"); // Add a div with class 'data-box'

client.println("<h2>Collected Data:</h2>");
client.println("<p><strong>Medication 1 Name:</strong> " + Med1Name + "<strong class='refill'> Refill: " +
Refill1 + "</strong></p>");
client.println("<p><strong>Medication 2 Name:</strong> " + Med2Name + "<strong class='refill'> Refill: " +
Refill2 + "</strong></p>");
client.println("<p><strong>Medication 3 Name:</strong> " + Med3Name + "<strong class='refill'> Refill: " +
Refill3 + "</strong></p>");
client.println("<p><strong>Medication 4 Name:</strong> " + Med4Name + "<strong class='refill'> Refill: " +
Refill4 + "</strong></p>");
client.print("<p><strong><style='font-size:34;'>Click <a href='/reset'>here</a> if you refill any
medication<br></strong></p>");

client.println("<br>"); // Add a line break here to separate the users
client.println("<br>"); // Add a line break here to separate the users
client.println("<br>"); // Add a line break here to separate the users
client.println("<p><strong>First User Name:</strong> " + User1Name + "</p>");
client.println("<p><strong>Number of Med from hoper 1 for User 1:</strong> " + String(Number1ofMed1) +
"</p>");
client.println("<p><strong>Number of Med from hoper 2 for User 1:</strong> " + String(Number1ofMed2) +
"</p>");
client.println("<p><strong>Number of Med from hoper 3 for User 1:</strong> " + String(Number1ofMed3) +
"</p>");
client.println("<p><strong>Number of Med from hoper 4 for User 1:</strong> " + String(Number1ofMed4) +
"</p>");

client.println("<br>"); // Add a line break here to separate the users
client.println("<br>"); // Add a line break here to separate the users
client.println("<br>"); // Add a line break here to separate the users
client.println("<p><strong>Second User Name:</strong> " + User2Name + "</p>");
client.println("<p><strong>Number of Med from hoper 1 for User 2:</strong> " + String(Number2ofMed1) +
"</p>");
client.println("<p><strong>Number of Med from hoper 2 for User 2:</strong> " + String(Number2ofMed2) +
"</p>");
client.println("<p><strong>Number of Med from hoper 3 for User 2:</strong> " + String(Number2ofMed3) +
"</p>");
client.println("<p><strong>Number of Med from hoper 4 for User 2:</strong> " + String(Number2ofMed4) +
"</p>");

client.println("</div>"); // Close the div
client.println("</body></html>");

```



```

*/

client.println("<html><head><style>");
client.println("body { background-color: White; background-size: cover; text-align: center; font-size: 34px; }");
client.println(".data-box { border: 2px solid black; padding: 20px; display: inline-block; }");
client.println(".medication-image { width: 600px; height: auto; }");
client.println("</style></head><body>");
client.println("<div class='data-box'>"); // Add a div with class 'data-box'

// Assuming you have URLs for the images of the medications
String med1ImageURL = "https://www.bhgrecovey.com/hubfs/BHG-New-Medication-5.8.13.jpg";
String med2ImageURL = "https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRmRWBEu-ORaMkwblh6LXFuwPcwr45tFpFb2w&s";
String med3ImageURL = "https://encrypted-tbn0.gstatic.com/images?q=tbn:ANd9GcRlk0KqhEi5VQgvzrotyh8MFf55Zh-XJ1w3aw&s";

client.println("<img src='\" + med1ImageURL + "\" class='\"medication-image\" alt='\"Medication 1 Image\"'>");
client.println("<h2>Collected Data:</h2>");
client.println("<br>"); // Add a line break here to separate the users
client.println("<br>"); // Add a line break here to separate the users
client.println("<br>"); // Add a line break here to separate the users

client.println("<p><strong>Medication 1 Name:</strong> " + Med1Name + "<strong class='\"refill\"'> Refill: " + Refill1 + "</strong></p>");
client.println("<p><strong>Medication 2 Name:</strong> " + Med2Name + "<strong class='\"refill\"'> Refill: " + Refill2 + "</strong></p>");
client.println("<p><strong>Medication 3 Name:</strong> " + Med3Name + "<strong class='\"refill\"'> Refill: " + Refill3 + "</strong></p>");
client.println("<p><strong>Medication 4 Name:</strong> " + Med4Name + "<strong class='\"refill\"'> Refill: " + Refill4 + "</strong></p>");
client.print("<p style='\"font-size:34px; color:red; border:2px solid black; padding:10px; display:inline-block;'><strong>Click <a href='\"/reset\"'>here</a> if you refill any medication<br></strong></p>");

client.println("<br>"); // Add a line break here to separate the users
client.println("<br>"); // Add a line break here to separate the users
client.println("<br>"); // Add a line break here to separate the users
client.println("<img src='\" + med2ImageURL + "\" class='\"medication-image\" alt='\"Medication 2 Image\"'>");
client.println("<br>"); // Add a line break here to separate the users
client.println("<p><strong>First User Name:</strong> " + User1Name + "</p>");
//client.println("<p><strong>Frequency of dispensation per day:</strong> " + String(NumberTime1) + "</p>");
client.println("<p><strong>No More then 4 Pills per Hoper </strong></p>");
client.println("<p><strong>Number of Med from hoper 1:</strong> " + Med1Name + "<strong class='\"refill\"'> for User 1:</strong> " + String(Number1ofMed1) + "</p>");
client.println("<p><strong>Number of Med from hoper 2:</strong> " + Med2Name + "<strong class='\"refill\"'> for User 1:</strong> " + String(Number1ofMed2) + "</p>");

```

```

        client.println("<p><strong>Number of Med from hoper 3:</strong> " + Med3Name + "<strong class=\"refill\"> for
User 1:</strong> " + String(Number1ofMed3) + "</p>");
        client.println("<p><strong>Number of Med from hoper 4:</strong> " + Med4Name + "<strong class=\"refill\"> for
User 1:</strong> " + String(Number1ofMed4) + "</p>");

        client.println("<br>"); // Add a line break here to separate the users
        client.println("<br>"); // Add a line break here to separate the users
        client.println("<br>"); // Add a line break here to separate the users
        client.println("<img src=\"\" + med3ImageURL + \"\" class=\"medication-image\" alt=\"Medication 3 Image\">");
        client.println("<br>"); // Add a line break here to separate the users
        client.println("<p><strong>Second User Name:</strong> " + User2Name + "</p>");
        //client.println("<p><strong>Frequency of dispensation per day:</strong> " + String(NumberTime2) + "</p>");
        client.println("<p><strong>No More then 4 Pills per Hoper </strong></p>");
        client.println("<p><strong>Number of Med from hoper 1:</strong> " + Med1Name + "<strong class=\"refill\"> for
User 2:</strong> " + String(Number2ofMed1) + "</p>");
        client.println("<p><strong>Number of Med from hoper 2:</strong> " + Med2Name + "<strong class=\"refill\"> for
User 2:</strong> " + String(Number2ofMed2) + "</p>");
        client.println("<p><strong>Number of Med from hoper 3:</strong> " + Med3Name + "<strong class=\"refill\"> for
User 2:</strong> " + String(Number2ofMed3) + "</p>");
        client.println("<p><strong>Number of Med from hoper 4:</strong> " + Med4Name + "<strong class=\"refill\"> for
User 2:</strong> " + String(Number2ofMed4) + "</p>");

        client.println("</div>"); // Close the div
        client.println("</body></html>");

        // The HTTP response ends with another blank line:
        client.println();
        // break out of the while loop:
        break;
    } else { // if you got a newline, then clear currentLine:
        currentLine = "";
    }
} else if (c != '\r') { // if you got anything else but a carriage return character,
    currentLine += c; // add it to the end of the currentLine
}

// Check to see if the client request was "GET /H" or "GET /L":
if (currentLine.endsWith("GET /reset")) {
    resetActions(); // Reset actions
    currentLine.endsWith("GET /");
    return;
}
}
}

// close the connection:

```

```

    client.stop();
    Serial.println("client disconnected");
}

}

void drawTextBox() {
    tft.fillRect(0, 0, 160, 40, ILI9341_BLACK);
    tft.drawRect(0, 0, 160, 40, ILI9341_WHITE);
    tft.setCursor(5, 5);
    tft.setTextColor(ILI9341_WHITE);
    tft.setTextSize(2);
    tft.print(inputString);
}

void drawKeyboard() {
    for (int x = 0; x < KEYBOARD_WIDTH; x++) {
        for (int y = 0; y < KEYBOARD_HEIGHT; y++) {
            if (keys[y][x] == '<') {
                tft.fillRect(KEY_X + x * (KEY_W + KEY_MARGIN), KEY_Y + y * (KEY_H + KEY_MARGIN), KEY_W + 30,
KEY_H, ILI9341_RED);
                tft.drawRect(KEY_X + x * (KEY_W + KEY_MARGIN), KEY_Y + y * (KEY_H + KEY_MARGIN), KEY_W + 30,
KEY_H, ILI9341_WHITE);
                tft.setCursor(KEY_X + x * (KEY_W + KEY_MARGIN) + 4, KEY_Y + y * (KEY_H + KEY_MARGIN) + 5);
                tft.setTextColor(ILI9341_WHITE);
                tft.setTextSize(2);
                tft.print("Back");
                y += 3;
            }
            else if (keys[y][x] == ' '){
                tft.fillRect(KEY_X + x * (KEY_W + KEY_MARGIN), KEY_Y + y * (KEY_H + KEY_MARGIN), KEY_W + 35,
KEY_H, ILI9341_GREEN);
                tft.drawRect(KEY_X + x * (KEY_W + KEY_MARGIN), KEY_Y + y * (KEY_H + KEY_MARGIN), KEY_W + 35,
KEY_H, ILI9341_WHITE);
                tft.setCursor(KEY_X + x * (KEY_W + KEY_MARGIN) + 4, KEY_Y + y * (KEY_H + KEY_MARGIN) + 5);
                tft.setTextColor(ILI9341_WHITE);
                tft.setTextSize(2);
                tft.print("Space");
                y += 3;
            }
            else if (keys[y][x] == '-') {
            }
            else {
                tft.fillRect(KEY_X + x * (KEY_W + KEY_MARGIN), KEY_Y + y * (KEY_H + KEY_MARGIN), KEY_W,
KEY_H, ILI9341_ORANGE);
            }
        }
    }
}

```

```

        tft.drawRect(KEY_X + x * (KEY_W + KEY_MARGIN), KEY_Y + y * (KEY_H + KEY_MARGIN), KEY_W,
KEY_H, ILI9341_WHITE);
        tft.setCursor(KEY_X + x * (KEY_W + KEY_MARGIN) + 8, KEY_Y + y * (KEY_H + KEY_MARGIN) + 5);
        tft.setTextColor(ILI9341_WHITE);
        tft.setTextSize(2);
        tft.print(keys[y][x]);
    }
}
}

void drawEnterButton() {
    tft.fillRect(170, 0, 65, 40, ILI9341_BLUE);
    tft.drawRect(170, 0, 65, 40, ILI9341_WHITE);
    tft.setCursor(175,5);
    tft.setTextColor(ILI9341_WHITE);
    tft.setTextSize(2);
    tft.print("Enter");
}

String getEnteredText() {
    return inputString;
}

void check() {
    int detected1 = digitalRead(DETECT1); // Read Laser sensor
    Serial.print("Sensor 1: ");
    Serial.println(detected1);
    Refill1 = skipSensor1 ? "Yes" : handleAction(detected1, ACTION1, startTime1, action1On, waitForReset1);
    if (Refill1 == "Yes") skipSensor1 = true;

    int detected2 = digitalRead(DETECT2); // Read Laser sensor
    Serial.print("Sensor 2: ");
    Serial.println(detected2);
    Refill2 = skipSensor2 ? "Yes" : handleAction(detected2, ACTION2, startTime2, action2On, waitForReset2);
    if (Refill2 == "Yes") skipSensor2 = true;

    int detected3 = digitalRead(DETECT3); // Read Laser sensor
    Serial.print("Sensor 3: ");
    Serial.println(detected3);
    Refill3 = skipSensor3 ? "Yes" : handleAction(detected3, ACTION3, startTime3, action3On, waitForReset3);
    if (Refill3 == "Yes") skipSensor3 = true;

    int detected4 = digitalRead(DETECT4); // Read Laser sensor
    Serial.print("Sensor 4: ");
    Serial.println(detected4);
    Refill4 = skipSensor4 ? "Yes" : handleAction(detected4, ACTION4, startTime4, action4On, waitForReset4);
}

```

```

if (Refill4 == "Yes") skipSensor4 = true;

// Now you can use Refill1, Refill2, Refill3, and Refill4 as needed
Serial.println("Refill status:");
Serial.println("Sensor 1: " + Refill1);
Serial.println("Sensor 2: " + Refill2);
Serial.println("Sensor 3: " + Refill3);
Serial.println("Sensor 4: " + Refill4);

delay(100); // Optional delay for debouncing
}

String handleAction(int detected, int actionPin, unsigned long &startTime, bool &actionOn, bool &waitForReset) {
    unsigned long currentTime = millis();
    unsigned long timeout = 60000; // 1 minute timeout

    if (waitForReset) {
        // Waiting for one false and one true detection
        if (detected == HIGH) {
            waitForReset = false; // Reset the wait flag after detecting HIGH
            actionOn = false; // Ensure action is off
        }
        return "";
    }

    if (detected == LOW) {
        if (!actionOn) {
            startTime = currentTime; // Record the start time
            actionOn = true; // Set the flag to indicate the action is on
        }
        digitalWrite(actionPin, HIGH); // Set the action ON
        Serial.println("Detected!");
        delay(100); // Optional delay for debouncing
    } else {
        digitalWrite(actionPin, LOW); // Set the action OFF
        Serial.println("No laser");
        actionOn = false;
        delay(100); // Optional delay for debouncing
    }

    // Check if the action has been on for more than 1 minute
    if (actionOn && (currentTime - startTime >= timeout)) {
        digitalWrite(actionPin, LOW); // Turn off the action
        Serial.println("Turning off action after 1 minute.");
        actionOn = false;
        waitForReset = true; // Set the wait flag to wait for reset condition
        delay(100); // Optional delay for debouncing
        return "Yes";
    }
}

```

```

    } else {
        return "No";
    }
}

void resetActions() {
    digitalWrite(ACTION1, LOW);
    digitalWrite(ACTION2, LOW);
    digitalWrite(ACTION3, LOW);
    digitalWrite(ACTION4, LOW);

    action1On = false;
    action2On = false;
    action3On = false;
    action4On = false;

    waitForReset1 = false;
    waitForReset2 = false;
    waitForReset3 = false;
    waitForReset4 = false;

    Refill1 = "";
    Refill2 = "";
    Refill3 = "";
    Refill4 = "";
    skipSensor1 = false;
    skipSensor2 = false;
    skipSensor3 = false;
    skipSensor4 = false;
    Serial.println("Reset button pressed. Variables reset.");
}

void sendDataToArduino() {
    sendDataToSlave(Number2ofMed1);
    delay(100);
    sendDataToSlave(Number2ofMed2);
    delay(100);
    sendDataToSlave(Number2ofMed3);
    delay(100);
    sendDataToSlave(Number2ofMed4);
    delay(100);
    sendDataToSlave(Number1ofMed1);
    delay(100);
    sendDataToSlave(Number1ofMed2);
    delay(100);
    sendDataToSlave(Number1ofMed3);
    delay(100);
}

```

```

sendDataToSlave(Number1ofMed4);
delay(100);
//sendDataToSlave(NumberTime1);
//delay(100);
//sendDataToSlave(NumberTime2);
//delay(100);
}

void sendDataToSlave(int data) {
    Wire.beginTransaction(9); // transmit to device #9
    Wire.write(data);
    Wire.endTransmission();
}

void loadImage(const char *filename) {
    ImageReturnCode stat; // Status from image-reading functions

    // Clear screen
    tft.fillScreen(ILI9341_BLACK);

    // Load and display image
    Serial.print(F("Loading "));
    Serial.print(filename);
    Serial.println(F(" to screen..."));
    stat = reader.drawBMP(filename, tft, 0, 0);
    reader.printStatus(stat);

    delay(350); // Pause 2 seconds before returning to loop()
}

String getUserInput() {
    inputString = "";
    bool userEntered = false;
    tft.fillScreen(ILI9341_BLACK);
    initKeyboard();
    while (!userEntered) {
        // Get touchscreen coordinates
        if (ts.touched()) {
            // Read touch point
            TS_Point p = ts.getPoint();
            // Scale touch coordinates to match the screen resolution
            p.x = map(p.x, 0, 240, tft.width(), 0);
            p.y = map(p.y, 0, 320, tft.height(), 0);

            int keyX = (p.x - KEY_X) / (KEY_W + KEY_MARGIN);
            int keyY = (p.y - KEY_Y) / (KEY_H + KEY_MARGIN);

            if (keyX >= 0 && keyX < KEYBOARD_WIDTH && keyY >= 0 && keyY < KEYBOARD_HEIGHT) {

```

```

char key = keys[keyY][keyX];
unsigned long currentMillis = millis();

if (currentMillis - lastKeyPress >= debounceDelay) {
    if (key == ' ')
        inputString += " ";
    else if (key == '<') {
        // Backspace key pressed
        if (inputString.length() > 0)
            inputString.remove(inputString.length() - 1); // Remove the last character
    }
    else
        inputString += key;

    drawTextBox();
    lastKeyPress = currentMillis;
    if (key == ' ')
        delay(500);
}
}
else if (p.x > 180 && p.x < 240 && p.y > 0 && p.y < 180) {
    // Enter button pressed
    Serial.println(inputString); // Print the entered text

    userEntered = true; // Set the flag to indicate that the user name has been entered
}
}
}
return inputString; // Return the entered string
}

void initKeyboard() {
    drawTextBox();
    drawKeyboard();
    drawEnterButton();
}

void printWifiStatus() {
    // print the SSID of the network you're attached to:
    Serial.print("SSID: ");
    Serial.println(WiFi.SSID());

    // print your board's IP address:
    IPAddress ip = WiFi.localIP();
    Serial.print("IP Address: ");
    Serial.println(ip);

    // print the received signal strength:

```



```
long rssi = WiFi.RSSI();  
Serial.print("signal strength (RSSI):");  
Serial.print(rssi);  
Serial.println(" dBm");  
// print where to go in a browser:  
Serial.print("To see this page in action, open a browser to http://");  
Serial.println(ip);  
}
```

APPENDIX B. Arduino Code for Project Follower code:

```
#include <Wire.h>
#include <SPI.h>
#include <MFRC522.h>
#include <Servo.h>

#define SLAVE_ADDRESS 9 // Address of the slave Arduino
#define RST_PIN 9 // Configurable, see typical pin layout above
#define SS_PIN 10 // Configurable, see typical pin layout above
#define SERVO_PIN_1 A0 // Servo 1 control pin
#define SERVO_PIN_2 A1 // Servo 2 control pin
#define SERVO_PIN_3 2 // Servo 3 control pin
#define SERVO_PIN_4 3 // Servo 4 control pin
#define SERVO_PIN_5 4 // Servo 5 control pin
#define SERVO_PIN_6 5 // Servo 6 control pin
#define SERVO_PIN_7 6 // Servo 7 control pin
#define SERVO_PIN_8 7 // Servo 8 control pin
#define DETECT1 8 // pin 8 for sensor

// Global variables to store received data
int Number2ofMed1 = 0;
int Number2ofMed2 = 0;
int Number2ofMed3 = 0;
int Number2ofMed4 = 0;
int Number1ofMed1 = 0;
int Number1ofMed2 = 0;
int Number1ofMed3 = 0;
int Number1ofMed4 = 0;
//int NumberTime1 = 0;
//int NumberTime2 = 0;

bool dataReceived = false; // Flag to indicate if data has been received
int receivedCount = 0; // Counter to keep track of received data count
bool firstNumberSkipped = false; // Flag to indicate if the first number has been skipped

MFRC522 mfrc522(SS_PIN, RST_PIN); // Create MFRC522 instance

Servo servo1;
Servo servo2;
Servo servo3;
Servo servo4;
Servo servo5;
Servo servo6;
Servo servo7;
Servo servo8;

void setup() {
  Serial.begin(115200); // Initialize serial communications with the PC
```

```

Wire.begin(SLAVE_ADDRESS);    // join I2C bus with address #9
Wire.onReceive(receiveEvent); // register event

// Wait until data is received
while (!dataReceived) {
    delay(1000);    // Check every second if data has been received
}

// Initialize SPI bus and MFRC522
SPI.begin();
mfrc522.PCD_Init();
delay(4);    // Optional delay. Some boards need more time after init to be ready
mfrc522.PCD_DumpVersionToSerial(); // Show details of PCD - MFRC522 Card Reader details
Serial.println(F("Scan PICC to see UID, SAK, type, and data blocks..."));

// Initialize servos
servo1.attach(SERVO_PIN_1);
servo2.attach(SERVO_PIN_2);
servo3.attach(SERVO_PIN_3);
servo4.attach(SERVO_PIN_4);
servo5.attach(SERVO_PIN_5);
servo6.attach(SERVO_PIN_6);
servo7.attach(SERVO_PIN_7);
servo8.attach(SERVO_PIN_8);

pinMode(DETECT1, INPUT);

// Set initial positions
servo1.write(70);
servo2.write(90);
servo3.write(165);
servo4.write(90);
servo5.write(90);
servo6.write(0);
servo7.write(160);
servo8.write(0);

delay(2000);

servo1.detach();
servo2.detach();
servo3.detach();
servo4.detach();
servo5.detach();
servo6.detach();
servo7.detach();
servo8.detach();
}

```

```

void loop() {
    // RFID card detection
    if (!mfrc522.PICC_IsNewCardPresent()) {
        return;
    }

    // Select one of the cards
    if (!mfrc522.PICC_ReadCardSerial()) {
        return;
    }

    // Check the UID of the detected card and perform corresponding actions
    int detected1 = digitalRead(DETECT1);
    if(detected1 == HIGH ){
        Serial.println("Ready");
        if (isTargetCard1()) {
            // Control servos for UID CC 62 54 49
            Serial.println("UID CC 62 54 49 detected:");
            controlServosBasedOnNumber1();
        } else if (isTargetCard2()) {
            // Control servos for UID 4F FC F1 1E
            Serial.println("UID 4F FC F1 1E detected:");
            controlServosBasedOnNumber2();
        }
    }

    // Dump debug info about the card; PICC_HaltA() is automatically called
    mfrc522.PICC_DumpToSerial(&(mfrc522.uid));
}

bool isTargetCard1() {
    byte targetUID1[] = {0xCC, 0x62, 0x54, 0x49};
    if (mfrc522.uid.size != 4) {
        return false;
    }
    for (byte i = 0; i < 4; i++) {
        if (mfrc522.uid.uidByte[i] != targetUID1[i]) {
            return false;
        }
    }
    return true;
}

bool isTargetCard2() {
    byte targetUID2[] = {0x4F, 0xFC, 0xF1, 0x1E};
    if (mfrc522.uid.size != 4) {
        return false;
    }

```

```

}
for (byte i = 0; i < 4; i++) {
    if (mfrc522.uid.uidByte[i] != targetUID2[i]) {
        return false;
    }
}
return true;
}

void controlServosBasedOnNumber1() {
    // Open and close servos based on the number of times specified
    for (int i = 0; i < Number1ofMed1; i++) {
        servo1.attach(SERVO_PIN_1);
        Serial.println("Operating Servo 1 for Number1ofMed1");
        Serial.println("Opening servo...");
        servo1.write(0);      // Open servo to 90 degrees (adjust if necessary)
        delay(1000);          // Keep open for 1 second

        Serial.println("Closing servo...");
        servo1.write(70);     // Close servo to 0 degrees (adjust if necessary)
        delay(1000);          // Ensure there's enough delay to complete the motion

        servo2.attach(SERVO_PIN_2);
        Serial.println("Operating Servo 2 for Number1ofMed2");
        Serial.println("Opening servo...");
        servo2.write(0);      // Open servo to 90 degrees (adjust if necessary)
        delay(1000);          // Keep open for 1 second

        Serial.println("Closing servo...");
        servo2.write(90);     // Close servo to 0 degrees (adjust if necessary)
        delay(1000);          // Ensure there's enough delay to complete the motion
        servo1.detach();
        servo2.detach();
    }

    for (int i = 0; i < Number1ofMed2; i++) {
        servo3.attach(SERVO_PIN_3);
        Serial.println("Operating Servo 3 for Number1ofMed3");
        Serial.println("Opening servo...");
        servo3.write(90);     // Open servo to 90 degrees (adjust if necessary)
        delay(1000);          // Keep open for 1 second

        Serial.println("Closing servo...");
        servo3.write(165);    // Close servo to 0 degrees (adjust if necessary)
        delay(1000);          // Ensure there's enough delay to complete the motion
    }
}

```

```

servo4.attach(SERVO_PIN_4);
Serial.println("Operating Servo 4 for Number1ofMed4");
Serial.println("Opening servo...");
servo4.write(0);          // Open servo to 90 degrees (adjust if necessary)
delay(1000);              // Keep open for 1 second

Serial.println("Closing servo...");
servo4.write(90);         // Close servo to 0 degrees (adjust if necessary)
delay(1000);              // Ensure there's enough delay to complete the motion
servo3.detach();
servo4.detach();
}

for (int i = 0; i < Number1ofMed3; i++) {
    servo5.attach(SERVO_PIN_5);
    Serial.println("Operating Servo 5 for Number1ofMed3");
    Serial.println("Opening servo...");
    servo5.write(0);       // Open servo to 90 degrees (adjust if necessary)
    delay(1000);           // Keep open for 1 second

    Serial.println("Closing servo...");
    servo5.write(90);      // Close servo to 0 degrees (adjust if necessary)
    delay(1000);           // Ensure there's enough delay to complete the motion

    servo6.attach(SERVO_PIN_6);
    Serial.println("Operating Servo 6 for Number1ofMed4");
    Serial.println("Opening servo...");
    servo6.write(120);     // Open servo to 90 degrees (adjust if necessary)
    delay(1000);           // Keep open for 1 second

    Serial.println("Closing servo...");
    servo6.write(0);       // Close servo to 0 degrees (adjust if necessary)
    delay(1000);           // Ensure there's enough delay to complete the motion
    servo5.detach();
    servo6.detach();
}

for (int i = 0; i < Number1ofMed4; i++) {
    servo7.attach(SERVO_PIN_7);
    Serial.println("Operating Servo 7 for Number1ofMed3");
    Serial.println("Opening servo...");
    servo7.write(50);      // Open servo to 90 degrees (adjust if necessary)
    delay(1000);           // Keep open for 1 second

    Serial.println("Closing servo...");
    servo7.write(160);     // Close servo to 0 degrees (adjust if necessary)

```

```

delay(1000);          // Ensure there's enough delay to complete the motion

servo8.attach(SERVO_PIN_8);
Serial.println("Operating Servo 8 for Number1ofMed4");
Serial.println("Opening servo...");
servo8.write(90);      // Open servo to 90 degrees (adjust if necessary)
delay(1000);          // Keep open for 1 second

Serial.println("Closing servo...");
servo8.write(0);       // Close servo to 0 degrees (adjust if necessary)
delay(1000);          // Ensure there's enough delay to complete the motion
servo7.detach();
servo8.detach();

}
}

void controlServosBasedOnNumber2() {
    // Open and close servos based on the number of times specified
    for (int i = 0; i < Number2ofMed1; i++) {
        servo1.attach(SERVO_PIN_1);
        Serial.println("Operating Servo 1 for Number2ofMed1");
        Serial.println("Opening servo...");
        servo1.write(90); // Open servo to 90 degrees (adjust if necessary)
        delay(1000);      // Keep open for 1 second

        Serial.println("Closing servo...");
        servo1.write(70); // Close servo to 0 degrees (adjust if necessary)
        delay(1000);      // Ensure there's enough delay to complete the motion

        servo2.attach(SERVO_PIN_2);
        Serial.println("Operating Servo 2 for Number2ofMed2");
        Serial.println("Opening servo...");
        servo2.write(90); // Open servo to 90 degrees (adjust if necessary)
        delay(1000);      // Keep open for 1 second

        Serial.println("Closing servo...");
        servo2.write(90); // Close servo to 0 degrees (adjust if necessary)
        delay(1000);      // Ensure there's enough delay to complete the motion
        servo1.detach();
        servo2.detach();
    }

    for (int i = 0; i < Number2ofMed2; i++) {

        servo3.attach(SERVO_PIN_3);

```

```

Serial.println("Operating Servo 3 for Number2ofMed3");
Serial.println("Opening servo...");
servo3.write(90);          // Open servo to 90 degrees (adjust if necessary)
delay(1000);              // Keep open for 1 second

Serial.println("Closing servo...");
servo3.write(165);        // Close servo to 0 degrees (adjust if necessary)
delay(1000);              // Ensure there's enough delay to complete the motion

servo4.attach(SERVO_PIN_4);
Serial.println("Operating Servo 4 for Number2ofMed4");
Serial.println("Opening servo...");
servo4.write(0);          // Open servo to 90 degrees (adjust if necessary)
delay(1000);              // Keep open for 1 second

Serial.println("Closing servo...");
servo4.write(90);          // Close servo to 0 degrees (adjust if necessary)
delay(1000);              // Ensure there's enough delay to complete the motion
servo3.detach();
servo4.detach();
}

for (int i = 0; i < Number2ofMed3; i++) {
    servo5.attach(SERVO_PIN_5);
    Serial.println("Operating Servo 5 for Number2ofMed3");
    Serial.println("Opening servo...");
    servo5.write(0);        // Open servo to 90 degrees (adjust if necessary)
    delay(1000);            // Keep open for 1 second

    Serial.println("Closing servo...");
    servo5.write(90);        // Close servo to 0 degrees (adjust if necessary)
    delay(1000);            // Ensure there's enough delay to complete the motion

    servo6.attach(SERVO_PIN_6);
    Serial.println("Operating Servo 6 for Number2ofMed4");
    Serial.println("Opening servo...");
    servo6.write(120);       // Open servo to 90 degrees (adjust if necessary)
    delay(1000);            // Keep open for 1 second

    Serial.println("Closing servo...");
    servo6.write(0);         // Close servo to 0 degrees (adjust if necessary)
    delay(1000);            // Ensure there's enough delay to complete the motion
    servo5.detach();
    servo6.detach();
}

```



```

for (int i = 0; i < Number2ofMed4; i++) {
    servo7.attach(SERVO_PIN_7);
    Serial.println("Operating Servo 7 for Number2ofMed3");
    Serial.println("Opening servo...");
    servo7.write(50);      // Open servo to 90 degrees (adjust if necessary)
    delay(1000);          // Keep open for 1 second

    Serial.println("Closing servo...");
    servo7.write(160);     // Close servo to 0 degrees (adjust if necessary)
    delay(1000);          // Ensure there's enough delay to complete the motion

    servo8.attach(SERVO_PIN_8);
    Serial.println("Operating Servo 8 for Number2ofMed4");
    Serial.println("Opening servo...");
    servo8.write(90);      // Open servo to 90 degrees (adjust if necessary)
    delay(1000);          // Keep open for 1 second

    Serial.println("Closing servo...");
    servo8.write(0);       // Close servo to 0 degrees (adjust if necessary)
    delay(1000);          // Ensure there's enough delay to complete the motion
    servo7.detach();
    servo8.detach();
}
}

void receiveEvent(int howMany) {
    while (Wire.available()) {
        // If the first number hasn't been skipped, skip it
        if (!firstNumberSkipped) {
            Wire.read();      // Read and discard the first number
            firstNumberSkipped = true; // Set the flag to indicate the first number has been skipped
        } else {
            // Read data from the I2C bus and store in respective variables
            int receivedData = Wire.read();
            delay(100);        // Introduce delay to ensure correct reading

            // Assign received data to the appropriate variable based on the received count
            switch (receivedCount) {
                case 0: Number2ofMed1 = receivedData; break;
                case 1: Number2ofMed2 = receivedData; break;
                case 2: Number2ofMed3 = receivedData; break;
                case 3: Number2ofMed4 = receivedData; break;
                case 4: Number1ofMed1 = receivedData; break;
                case 5: Number1ofMed2 = receivedData; break;
                case 6: Number1ofMed3 = receivedData; break;
            }
        }
    }
}

```

```

        case 7: Number1ofMed4 = receivedData; break;
        //case 8: NumberTime1 = receivedData; break;
        //case 9: NumberTime2 = receivedData; break;
        default: break;
    }

    receivedCount++;          // Increment the received count

    // If all 8 data sets have been received, reset the receivedCount and firstNumberSkipped flags
    if (receivedCount >= 8) {
        dataReceived = true;    // Set the flag to indicate that data has been received
        receivedCount = 0;      // Reset the received count for next data set
        firstNumberSkipped = false; // Reset the first number skipped flag
    }
}
}
printReceivedData();
}

void printReceivedData() {
    Serial.println("Received Data:");
    Serial.print("Number2ofMed1: ");
    Serial.println(Number2ofMed1);
    Serial.print("Number2ofMed2: ");
    Serial.println(Number2ofMed2);
    Serial.print("Number2ofMed3: ");
    Serial.println(Number2ofMed3);
    Serial.print("Number2ofMed4: ");
    Serial.println(Number2ofMed4);
    Serial.print("Number1ofMed1: ");
    Serial.println(Number1ofMed1);
    Serial.print("Number1ofMed2: ");
    Serial.println(Number1ofMed2);
    Serial.print("Number1ofMed3: ");
    Serial.println(Number1ofMed3);
    Serial.print("Number1ofMed4: ");
    Serial.println(Number1ofMed4);
    //Serial.print("NumberTime1: ");
    //Serial.println(NumberTime1);
    //Serial.print("NumberTime2: ");
    //Serial.println(NumberTime2);
}

```

APPENDIX C. Arduino Code for Project: arduino_secrets.h

```
#ifndef ARDUINO_SECRETS_H
#define ARDUINO_SECRETS_H

#define SECRET_SSID "JaysWiFi" // Replace with your iPhone's hotspot name
#define SECRET_PASS "Jaypatel4502" // Replace with your iPhone's hotspot password

#endif // ARDUINO_SECRETS_H
```

APPENDIX D. Code of Ethics for Engineering: IEEE and NSPE

IEEE Code of Ethics

- I. To uphold the highest standards of integrity, responsible behavior, and ethical conduct in professional activities.
 1. To hold paramount the safety, health, and welfare of the public, to strive to comply with ethical design and sustainable development practices, to protect the privacy of others, and to disclose promptly factors that might endanger the public or the environment;
 2. To improve the understanding by individuals and society of the capabilities and societal implications of conventional and emerging technologies, including intelligent systems;
 3. To avoid real or perceived conflicts of interest whenever possible, and to disclose them to affected parties when they do exist;
 4. To avoid unlawful conduct in professional activities, and to reject bribery in all its forms;
 5. To seek, accept, and offer honest criticism of technical work, to acknowledge and correct errors, to be honest and realistic in stating claims or estimates based on available data, and to credit properly the contributions of others;
 6. To maintain and improve our technical competence and to undertake technological tasks for others only if qualified by training or experience, or after full disclosure of pertinent limitations;
- II. To treat all persons fairly and with respect, to not engage in harassment or discrimination, and to avoid injuring others.
 7. To treat all persons fairly and with respect, and to not engage in discrimination based on characteristics such as race, religion, gender, disability, age, national origin, sexual orientation, gender identity, or gender expression;
 8. To not engage in harassment of any kind, including sexual harassment or bullying behavior;
 9. To avoid injuring others, their property, reputation, or employment by false or malicious actions, rumors or any other verbal or physical abuses;
- III. To strive to ensure this code is upheld by colleagues and co-workers.

10. To support colleagues and co-workers in following this code of ethics, to strive to ensure the code is upheld, and to not retaliate against individuals reporting a violation.

NSPE Code of Ethics for Engineers

The National Society of Professional engineers (NSPE) have developed a more extensive code of ethics that is of common use in industry. The link to the NSPE Code of Ethics is:

<http://www.nspe.org/resources/ethics/code-ethics>

APPENDIX E. User Manual

Getting Started

1. **Plug in the device** to a standard three-prong electrical outlet.
2. Once connected, the system will automatically power on, and the PillPros application will start on the touchscreen display.
3. While the application is loading, **begin loading the medications into the pill bins:**
 - a. Place **only one type of medication** per pill bay.
 - b. **Spread the pills evenly** within the bay to ensure proper dispensing.
 - c. Note which medication is placed in each bay, as the bays are **numbered** for reference.

1. Setting Up a New User

The PillPros touchscreen interface allows users to quickly and accurately set up their personalized medication schedules. Follow these steps to set up a user:

Step 1: Add a New User

1. Tap the "**Add User**" button.
2. Enter the user's name.

Step 2: Enter Dosages

1. The system will prompt you to enter the dosage for **Pill Bay 1**.
 - a. Use the touchscreen keypad to enter the number of pills to be dispensed from Bay 1.
 - b. Confirm the dosage by tapping the "**Enter**" button.
2. Repeat this step for **Pill Bays 2, 3, and 4**, entering the dosage for each bay.

Note: Be sure to review and confirm each entry for accuracy.

3. If additional users need to be added, repeat **Step 1 through Step 3** for each new user.

2. Enter Medications

- Tap the "**Medication Setup**" icon on the touchscreen.
- The system will prompt you to **enter the name of the medication** for **Pill Bay 1**.
- Use the on-screen keyboard to type the medication name and confirm.
- Repeat this process for **Pill Bays 2, 3, and 4**, entering the name of the medication assigned to each bay.
- After entering the medication names, confirm the setup

3. Tracking Medication Schedules

The PillPros system integrates with a smartphone app to provide users and caregivers with real-time access to medication schedules and dosage history.

1. Open the PillPros mobile app.
2. View and track:
 - a. Users
 - b. Medication schedules.
 - c. Dosage history.
 - d. Notifications for any adjustments or refills needed.

This feature allows users to stay informed and make adjustments as required.

4. Dispensing Medication

The PillPros system dispenses medication accurately based on the user's setup. Follow these steps to dispense medication:

1. **Insert a cup** into the designated dispensing area of the machine.
2. **Press the user's RFID tag** against the scanner located on the device.
3. Once the RFID tag is recognized, the system will automatically begin dispensing the medication.

Note: If a cup is not inserted, the machine will not start dispensing to prevent pills from being misplaced or lost.

Example 1: Dispensing Medication for June

Based on June's setup, the system dispenses:

- **2 pills of Advil** from Hopper 1.
- **1 pill of Naran** from Hopper 2.
- **3 pills of Tesla** from Hopper 3.
- **4 pills of Qubic** from Hopper 4.

Example 2: Dispensing Medication for May

Based on May's setup, the system dispenses:

- **3 pills of Advil** from Hopper 1.
- **2 pills of Naran** from Hopper 2.
- **4 pills of Tesla** from Hopper 3.
- **1 pill of Qubic** from Hopper 4.

The system ensures precise and reliable dispensing, delivering the exact dosage for each user.